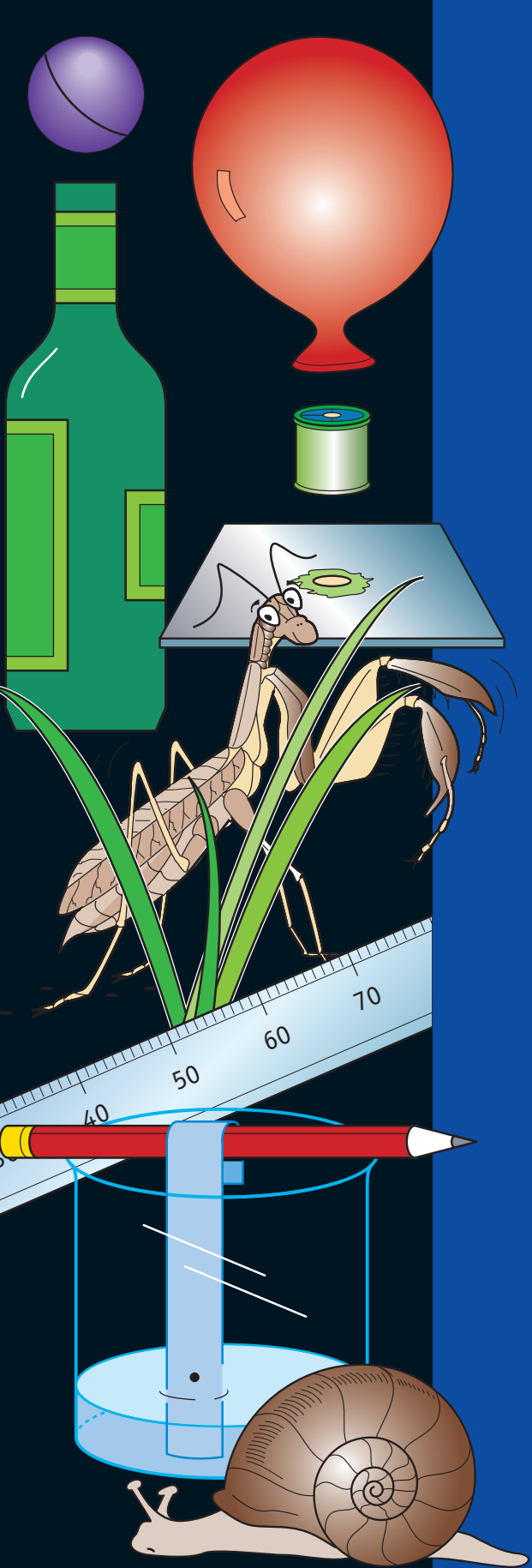




# Attention Grabbing Science

Book 1

David Palmer

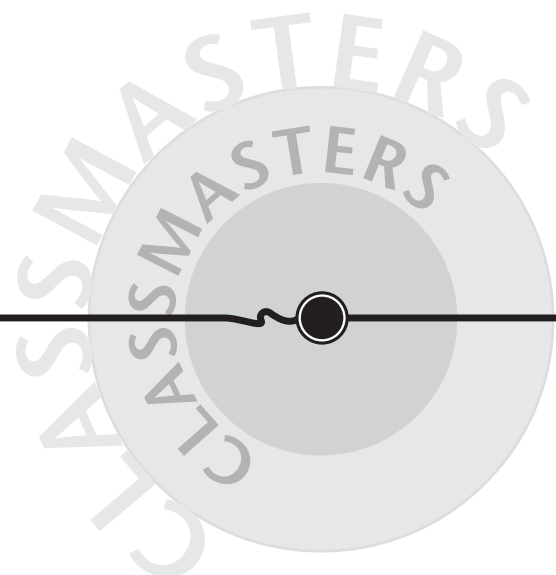


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# **Attention Grabbing Science**

**Book 1**

**David Palmer**





# ACKNOWLEDGMENTS

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I would like to thank Julie, Owen and Patrick for helping me choose the activities, and for letting me work on the weekends.

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# PREFACE

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Professor Julius Sumner Miller showed that science activities that have the element of surprise (that is, an unexpected result) are interesting to people of all ages. A surprising result can motivate students to examine their own ideas and to want to know 'Why is it so?'. With similar thinking, I decided to base this book on a collection of surprising science activities.

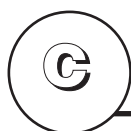
However, learning science is not just about whiz-bang activities, so this book also contains lots of suggestions for other types of motivational science experiences. These have been grouped together into themes so that students can experience a variety of activities in relation to each concept. The 'To Investigate' sections have ideas for student-designed investigations. The shaded boxes contain fascinating facts chosen for their surprise value as well as interesting historical anecdotes. The 'Linked Activities' sections contain further surprising activities as well as ideas for problem solving, dramatisation, modelling and direct experience of phenomena. Finally, the 'Creative Literacy' boxes contain motivational ways for students to record and express their knowledge.

This book is intended to encourage motivation in science, but it is not about big bangs and explosions nor does it emphasise the use of scientific apparatus. In fact, a conscious decision was made to include many activities that do not rely on specialised equipment. My belief is that if science is something that involves everyday materials (and activities that can be done at home in the kitchen) then this automatically makes it relevant to real life.

An important criterion for the selection of activities was that they can all be relied on to work with a minimum of fuss. However, like all science activities, they are not completely foolproof. Readers would be advised to abide by Dave's Golden Rule, which states 'Always, always, always, always, always, test it by yourself first, before you try to impress other people'. In this way you can check it with the materials you have available and adapt it for your conditions.

There are detailed safety instructions for any experiments that might involve risk factors. Please note, however, that this book is designed for use by teachers in their classrooms, not for students in an unsupervised environment. Teachers should follow all normal safety procedures.

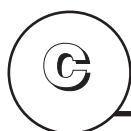
One issue that became apparent to me when I first began writing this book was that many of the activities relate to physical science concepts and that, by comparison, biology and chemistry are less well represented. However, I make no apology for this. I believe that different sciences are interesting for different reasons, and one of the features of physics is that it does lend itself to the creation of surprising hands-on activities. In fact, with recent reports that students' interest in physics is falling (especially among girls), it could be argued that it is in precisely this area that we need to develop interesting and motivational experiences.



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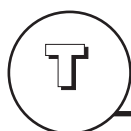
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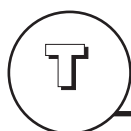
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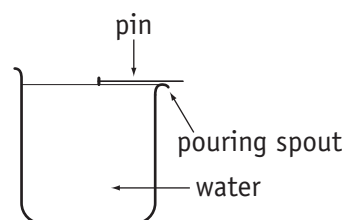
## 1

# FLOAT A PIN

You can make a metal pin stay on top of the water.

## HOW TO DO IT

- 1 Fill a beaker with water up to its pouring spout.
- 2 At the pouring spout, gently place the head of a dry, clean pin onto the surface of the water.
- 3 Give the sharp end a gentle tap with your fingernail so the pin slides out onto the surface of the water. (*The pin will remain on top of the water.*)



## EXPLANATION

The force of attraction (cohesive force) between the molecules at the surface of the water is strong enough to support a pin.

## TO INVESTIGATE

Do other liquids have surface tension? Try salt water, warm water, cooking oil and methylated spirits.

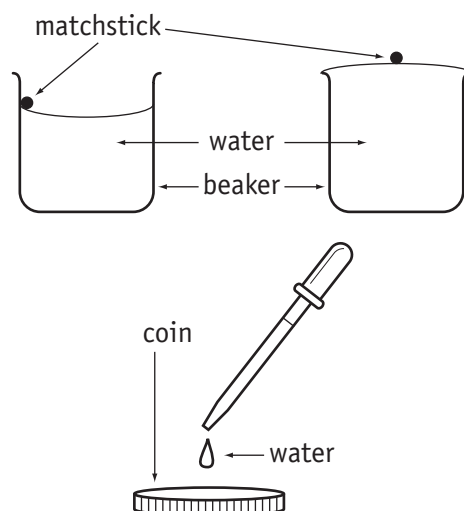
*Safety warning: be careful of methylated spirits' fumes. (Surface tension is usually lower in warm water and organic liquids.)*

**The pin in the activity above was not wet. It was not actually *in* the water. It was on *top* of the water and was therefore still dry.**

**CREATIVE LITERACY** Is this why steel ships float on water? Contribute your argument to a class discussion on this question.

## LINKED ACTIVITIES

- 1 **Confuse a matchstick.** Half fill a small (100 mL) beaker with water. Drop a small piece of match stick (5 mm long) into the middle of the water. (*It will move towards the glass because the meniscus curves upwards there, and the stick floats to the highest point of the water.*) Now fill the beaker above its rim and float the stick next to the glass. (*It will move away from the glass because surface tension holds the water in, and the highest point of the water is now in the middle.*)
- 2 **Water a coin.** Put a single drop of water on a 5c coin and use it to predict how many drops of water you could fit on the coin. Use a pipette to administer 1 drop at a time until the water overflows. (*Surface tension will hold more water on the coin than most people predict.*)



**Some animals can walk on water. Water striders (*Gerris*) are insects with hairs on their feet to spread their weight so they do not penetrate the surface of the water. They prey upon other insects that have become trapped in the water.**

**CREATIVE LITERACY** Could this explain the story of Jesus walking on water? Write a paragraph which explains your arguments as to whether a person could walk on water.

## 2

## WATER IN A SIEVE

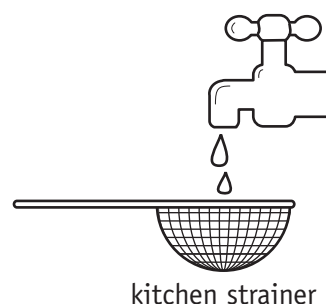
Wire mesh can hold water.

### HOW TO DO IT

Hold a dry wire kitchen strainer over a sink. Carefully put 2 or 3 drops of water into it. (*The water will stay in the strainer.*)

### EXPLANATION

Water is held in the mesh by adhesive forces between the wire and water and cohesive forces within the water. A layer of water forms in each hole in the mesh and the surface tension stops the rest of the water from passing through.



### TO INVESTIGATE

What would happen if you put some detergent in the water? What would happen if you used oil or methylated spirits instead of water? (*Safety warning: be careful of methylated spirits' fumes.*)

If you hold a piece of canvas up to the light, you can see that it has lots of tiny holes in it, yet it still manages to keep the rain out. When it rains, the first drops of water seal the holes, so the rest of the water runs off. But the Golden Rule of Canvas Tents is: don't touch the inside of the tent when it is raining. If you do, adhesive forces between the water and your finger will drag the water into the tent.

#### CREATIVE LITERACY

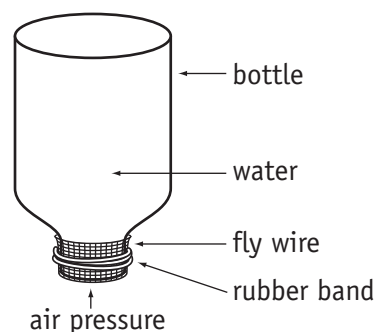
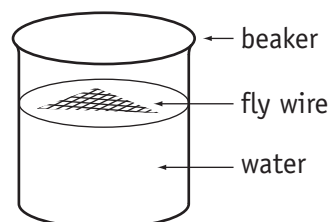
The saying 'coming home with a wet sail' is popular with sports commentators. Have a class discussion about the link between science and sport here.

#### HISTORICAL NOTE

The word 'canvas' is derived from 'cannabis'. Early types of sail canvas were often made from hemp (cannabis).

### LINKED ACTIVITIES

- 1 Make a raft from fly wire.** Cut out a triangular shape from a sheet of fly wire. Gently lower it onto the surface of the water in a beaker. (*The raft will float because the water seals all the holes. You can observe this by lifting it out of the water.*) Add a few drops of detergent to the water. (*The raft will sink because the cohesive forces have been weakened by the detergent.*)
- 2 Make a fly wire bottle top.** Cover the mouth of a bottle with a layer of fly wire and fasten it with a rubber band. Fill the bottle with water. (*The water will go straight through the fly wire.*) Quickly tip the bottle upside down. (*The water will not run out because the water has sealed the holes in the fly wire and air pressure exerts a force that counters the weight of the water.*)



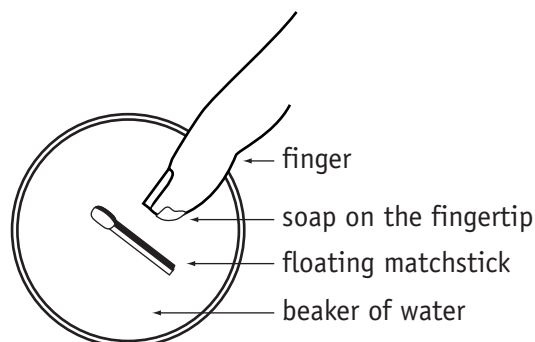
## 3

# MAGIC FINGER

A floating match will suddenly spring away from your finger.

## HOW TO DO IT

- 1 Fill a beaker with water and gently place half a matchstick onto the surface of the water. (*You will be able to see water adhering to the matchstick.*)
- 2 Put a small drop of liquid soap on the tip of your finger and touch this 'magic finger' to the surface of the water to one side of the match. (*The matchstick will suddenly spring away from your finger.*)



## EXPLANATION

This activity demonstrates the tension in surface tension. The soap weakened the surface tension on one side of the match. The match sprang away because the surface tension on the other side was stronger (like a wire under tension will spring away from the point where you cut it).

## TO INVESTIGATE

Is it possible to repeat this activity using the same water? (*No, you will need to use fresh water and a new match and remove all the soap in the beaker before you can repeat the activity.*)

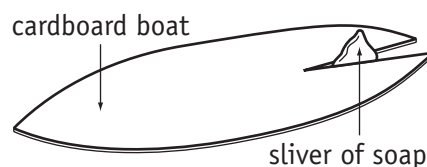
**Some beetles (*Stenus*) walk slowly on the surface of mountain streams. But when threatened, they dab the end of their abdomen on the water and release a chemical that lowers the surface tension so dramatically they are whisked away almost faster than the eye can see.**

### HISTORICAL NOTE

Soap wasn't used very much in the Middle Ages. In fact, a parcel of new soap was accompanied by detailed instructions for how to use it.

## LINKED ACTIVITIES

- 1 **Make a soap-powered boat.** Cut out a small boat shape from light cardboard. Cut a notch in the back and fit a small flake of soap into it, then put the boat in fresh tap water. (*The boat will move forward because the surface tension is lowered at the back.*)
- 2 **Design a big-bubble recipe.** Try mixing different ratios of dishwashing liquid, water and either sugar or glycerine or gelatine. (*When surface tension is weakened, the distance between the water molecules can be increased to form bubbles. Substances such as sugar, glycerine and gelatine help to make the bubbles last longer because they form weak adhesive bonds with water. This helps to slow down the rate of evaporation of the water from the large surface area of the bubbles. A good bubble mixture is one part dishwashing liquid, one part glycerine and three parts water.*)



**CREATIVE LITERACY** Write a list of instructions for your method of making big bubbles.

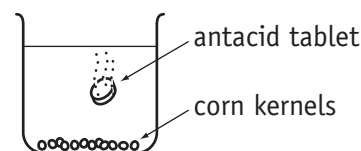
## 4

## CORN KERNELS

When corn kernels are put in fizzy water they go up and down.

## HOW TO DO IT

- 1 Put about a dozen kernels of popping corn in a 200 mL beaker of water. (*They will sink.*)
- 2 Put an antacid tablet in the water, wait for it to finish fizzing and then observe the corn. (*The corn kernels will rise to the surface and then sink many times over the next few minutes.*)



## EXPLANATION

The corn kernel by itself is more dense than water, so it sinks. However, the surface of the kernel provides sites for bubbles of carbon dioxide to form. The average density of the corn-bubble unit is lower than that of water, so it rises to the surface. The bubbles burst, and the process is repeated.

## TO INVESTIGATE

Does this phenomenon apply only to corn and antacid? Try 'hundreds and thousands' and currants; and try other types of carbonated drinks.

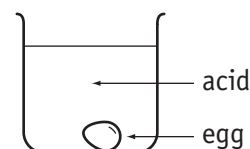
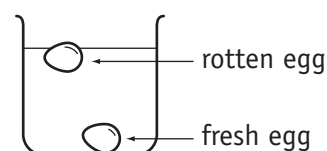
**Pumice is a rock that floats, and ebony is a wood that sinks.**

## HISTORICAL NOTE

According to popular belief, over 50 ships have disappeared in the Bermuda Triangle. One (unproven) theory is that methane gas bubbles up from the ocean floor and makes the water less dense, so a ship would sink very rapidly if it entered such an area.

## LINKED ACTIVITIES

- 1 **Do the rotten egg test.** Let an egg go rotten, then place it in a beaker of water with a fresh egg. (*The fresh egg will sink, but the rotten egg will float because gases are produced by bacteria as they decompose the contents of the egg.*)
- 2 **Make an egg float in acid.** *Safety glasses and laboratory coats should be worn.* Place a fresh egg in a beaker and add dilute hydrochloric acid until the beaker is almost full. (*The egg will initially sink, but will then rise to the surface as the acid reacts with the calcium carbonate in the shell of the egg to produce bubbles of carbon dioxide.*)



**CREATIVE LITERACY** Give a full verbal explanation of the rotten egg test to your parents. Ask them to comment on how well they comprehended what you were saying. Summarise their comments.

## 5

# CARTESIAN DIVER

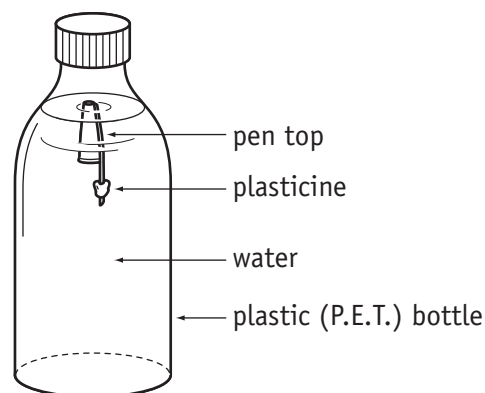
A floating pen top can be made to sink and then rise without being touched.

## HOW TO DO IT

- 1 Attach some plasticine to a plastic pen top, so that it only just floats in water. Be careful not to block the opening of the pen top.
- 2 Lower the pen top into a plastic bottle filled with water.
- 3 Screw the lid on tightly, then squeeze the sides of the bottle. *(The pen top will sink when you squeeze the bottle and rise again when you stop squeezing.)*

## EXPLANATION

When you squeeze the bottle the water transfers the pressure to the air inside the pen top. The air inside the pen top becomes compressed, so the unit (air pocket + pen top) becomes more dense than the water, and it sinks.



## TO INVESTIGATE

Can you find any evidence that the air inside the pen top becomes compressed when you squeeze the bottle? *(You can observe it if you look into the pen top from below.)*

**Sperm whales can hold their breath for almost 2 hours.**

**They can swim down to 1 km beneath the surface.**

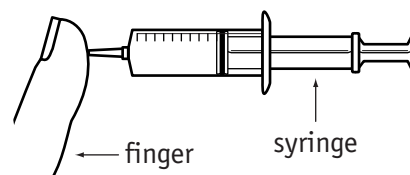
**Modern military submarines can circumnavigate the globe without surfacing.**

## HISTORICAL NOTE

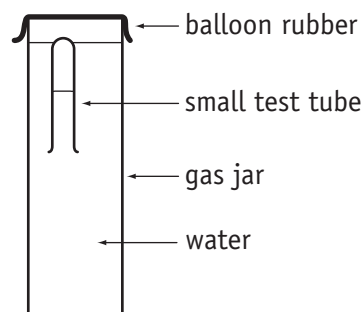
The first submarine, built in 1620, was covered with leather and powered by oars. It travelled up the Thames River and could remain at a depth of 5 m for about 15 minutes.

## LINKED ACTIVITIES

- 1 **Compress air.** Use a large plastic syringe to demonstrate that air can be compressed but not water. Use the gradations on the syringe to work out the percentage compression you can achieve.
- 2 **Solve the following problem.** How would you make a Cartesian diver using laboratory equipment? *(The diagram shows one alternative.)*



**CREATIVE LITERACY** Scuba divers breathe compressed air and may get 'the bends' if they come to the surface too quickly. Is opening a bottle of cola like a scuba diver getting the bends? Write your response to this question.



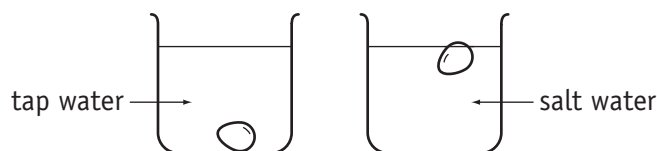
## 6

# FLOATING EGG

An egg will sink in one beaker of water but float in another.

## HOW TO DO IT

- 1 Gently place a fresh egg in a beaker of tap water. (*The egg will sink.*)
- 2 Transfer the egg to a beaker of very salty water. (*The egg will float.*)



## EXPLANATION

Salt water is more dense than fresh water. The buoyancy force (or upthrust) is greater in fluids of higher density. In salt water there is enough upthrust for the egg to float.

## TO INVESTIGATE

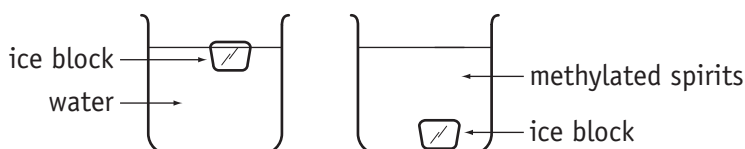
Predict how the weights of equal volumes of the salt water and tap water used in this activity would compare. Use a volumetric pipette to measure volumes and a digital balance to compare weights; calculate their densities.

**The oceans came from rocks. The space rocks that originally formed the earth contained small amounts of water chemically combined with minerals in the rocks. When the rocks melted in the interior of the earth, the water was released as steam, which emerged from volcanoes and eventually formed the oceans.**

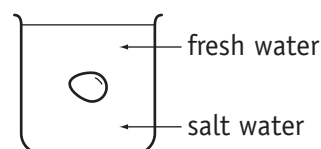
**When salt water freezes, the salt is forced out, so the ice is fresh water.**

## LINKED ACTIVITIES

- 1 **Make an ice block sink.** *TEACHER DEMONSTRATION ONLY* — Put one ice block in water and another one in methylated spirits. *Safety warning: be careful of fumes. (Water is more dense than methylated spirits, so the ice will float in the former and sink in the latter.)*



- 2 **Separate salt water from fresh water.** Half fill a beaker with very salty water then float an egg in it. Slowly and gently add fresh water to the side of the beaker until it is full. (*The two liquids do not fully mix, so the egg will remain halfway up the beaker at their interface.*)



**Some parts of the sea are fresh water. Fresh water and salt water don't automatically mix, so when the water from the Amazon River enters the Atlantic Ocean it continues as a body of fresh water for 100 km out to sea.**

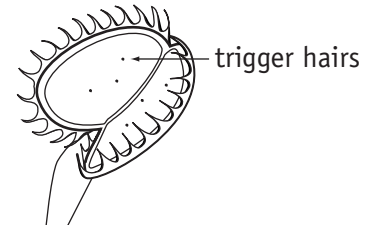
**CREATIVE LITERACY** Imagine that you are a ship's captain. You are out at sea and your ship is so full that it is only just floating. You are worried about an approaching storm, but the nearest port is several kilometres up a large river. Should you head for this port? Explain your reasons.

## VENUS FLYTRAP

This plant can move fast enough to catch insects.

### HOW TO DO IT

- 1 Buy a potted Venus flytrap (*Dionaea*) from a nursery.
- 2 Find a pale-coloured trigger hair on the inside surface of a trap and gently touch it with a pin. (*The trap will close.*)



### EXPLANATION

As venus flytraps are green plants, they produce their food by photosynthesis. However, they are often found growing in poor soils, so they use their insect prey as a supplementary source of minerals, especially iron and phosphorus compounds.

### TO INVESTIGATE

If a trap closes without having caught any prey, will it open up faster than one that has caught prey (such as a dead ant)?

**Plants make up about 90% of the mass of all living things on earth.**

**Most of the oxygen in the air is produced by microscopic algae in the oceans, rather than by land plants.**

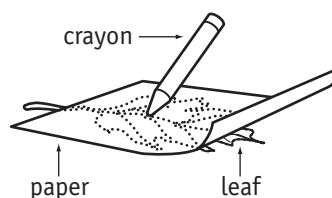
**The earth's forests are being cut down at a rate of about 40 hectares per minute. A lot of the wood is used to make toilet paper.**

#### HISTORICAL NOTE

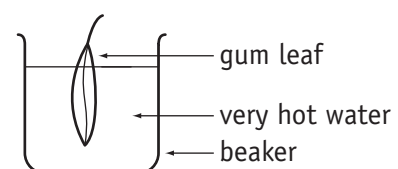
Charles Darwin considered the Venus flytrap to be 'the most wonderful plant in the world'.

### LINKED ACTIVITIES

- 1 **Make leaf rubbings** to record differences between dicotyledons and monocotyledons. Place a sheet of paper over the leaf and lightly colour the top of it with the side of a soft crayon. (*The shape of the leaf, and its veins, will become clearly visible.*)



- 2 **Locate the stomata.** *TEACHER DEMONSTRATION ONLY*  
Lower a freshly picked gum leaf into a beaker of water that has just boiled. (*The lower surface of the leaf will become covered with tiny bubbles, but the upper surface will have few or none. The air inside the leaf has expanded from the heat. Some of it appears outside the stomata, most of which are on the lower surface.*)

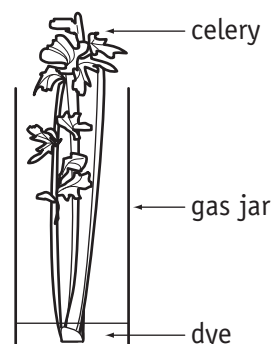


## CELERY STALKS

Water travels surprisingly quickly up a celery stalk.

### HOW TO DO IT

- 1 Take a gas jar and pour some red food colouring into it (a depth of 0.5 cm is enough). Do not dilute the food colouring.
- 2 Cut a fresh celery stalk with its leaves still attached and put the lower end of the stalk into the dye. Place it in a warm sunny position. Record the time.
- 3 Remove the stalk after 3 minutes and progressively cut up the stalk to see how far the dye has gone. (*The dye should have travelled right up the stem.*)



### EXPLANATION

The coloured water travels up the xylem vessels in the stalk by adhesive forces (between the water and the walls of the vessels) and cohesive forces (between the water molecules). Evaporation from the leaves maintains the flow.

**The Australian mountain ash (*Eucalyptus regnans*) is the tallest flowering plant in the world. Water travels upwards over 90 m in some of these trees.**

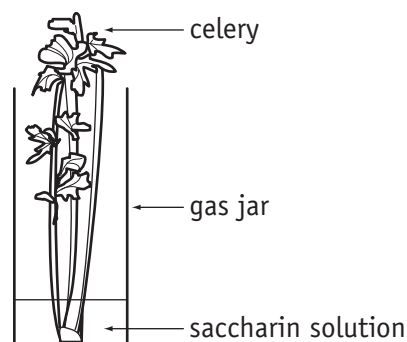
**CREATIVE LITERACY** Imagine that you are a molecule of water. Describe your experiences in a mountain ash tree.

### HISTORICAL NOTE

Near the town of Derby, in Western Australia, the hollow trunk of one baobab tree used to be a gaol. The opening was fitted with a metal grille, and up to 18 prisoners could be kept inside.

### LINKED ACTIVITIES

- 1 **Make diet celery.** Put about 10 saccharin tablets in 100 mL of water. Stand a celery stalk in the liquid and leave it overnight. Taste the celery. (*The saccharin will be tasted in the celery. Humans can taste saccharin at concentrations hundreds of times lower than sugar.*)
- 2 **Examine the climatic record of your local area.** Obtain a firewood log in which the growth rings are clearly visible. Count the rings and compare their widths. (*Many trees produce one growth ring each year. Wide rings indicate years of high rainfall or warm temperatures, with consequent greater growth.*)



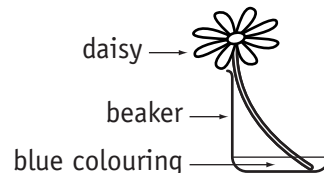
**You can eat rhubarb stems, but their leaves are poisonous.**

## DYEING DAISIES

You can make a daisy turn green.

### HOW TO DO IT

- 1 Obtain a yellow daisy and cut the stem so it is only 5–6 cm long.
- 2 Place the cut end of the stem in a small beaker containing some undiluted blue food colouring.
- 3 Put the beaker and flower in a warm sunny part of the room. *(After a few minutes the blue colour will begin to appear in the petals. As more dye enters the petals they will eventually turn green.)*



### EXPLANATION

The food colouring travels up the xylem vessels in the stem of the plant and enters the petals. The green colour results from the combination of the yellow petals and the blue dye.

### TO INVESTIGATE

Does this phenomenon apply only to daisies? Also, is it possible to colour a flower with two different colours?

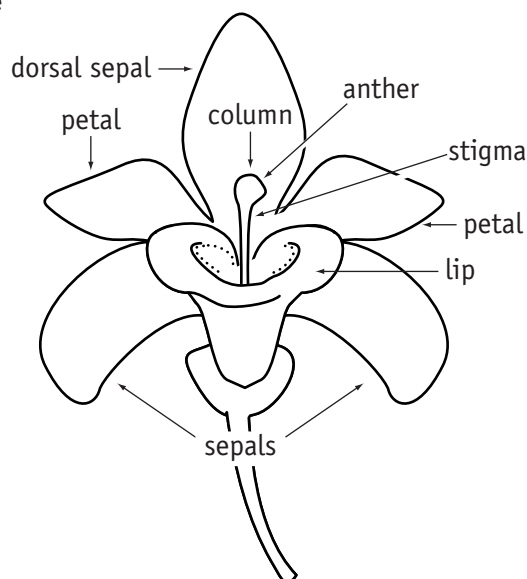
**Not all flowers are sweet smelling. The stinking corpse lily (*Rafflesia*), of Malaysian jungles, is the world's largest flower, growing up to 1 m in diameter. It makes a smell like rotting meat to attract flies to pollinate it.**

**Not all insect visitors at flowers are collecting nectar. Some orchids have their petals arranged in such a way as to resemble a female wasp. Male wasps are attracted to the flowers and try to mate with them, but they only succeed in pollinating the flowers.**

**CREATIVE LITERACY** Write a poem summarising the functions of the anther, filament, stigma, style, ovary, petals and nectaries.

### LINKED ACTIVITIES

- 1 **Dissect an orchid.** Obtain a large orchid and identify the features which make it different to other flowers. *(Orchid flowers are vertically symmetrical. The sepals are coloured, the lower petal forms a lip that is a landing platform for insects, and the male and female parts are combined into a column inside the flower. Any insect that lands on the lip and falls into the centre of the flower will brush against the column, resulting in pollination.)*
- 2 **Have a scavenger hunt** for plant reproductive structures that don't have petals. During spring, find the flowers of grasses, eucalypts and bottle brushes plus the reproductive structures of conifers, ferns and mosses.



### HISTORICAL NOTE

Tomatoes used to be only ornamental plants. People were too scared to eat them because they thought the fruit would be poisonous. In fact, the roots and leaves of the tomato ARE poisonous.

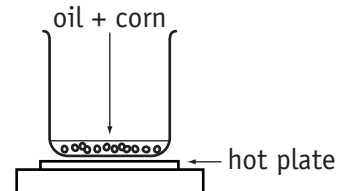
You can make corn seeds explode.

## HOW TO DO IT

### TEACHER DEMONSTRATION ONLY

*Safety glasses and laboratory coat should be worn.*

- 1 Obtain a large beaker and clean it thoroughly.
- 2 Cover the bottom of the beaker with about 4 tablespoons of cooking oil.
- 3 Place the beaker on a hotplate.
- 4 Add about  $\frac{1}{2}$  a cup of popping corn and  $\frac{1}{2}$  a teaspoon of salt, then cover the beaker.
- 5 Mix the contents regularly as it heats. *(After a few minutes the corn will pop.)*



## EXPLANATION

Corn seeds contain a small amount of water that is mixed with the starch. When this water is heated it eventually turns into steam, but the steam is trapped inside the seed by the hard outer coat. The steam pressure builds up until the kernel explodes, exposing the starch in a frothy state.

## TO INVESTIGATE

Do popping corn kernels germinate as readily as corn from seed packets?

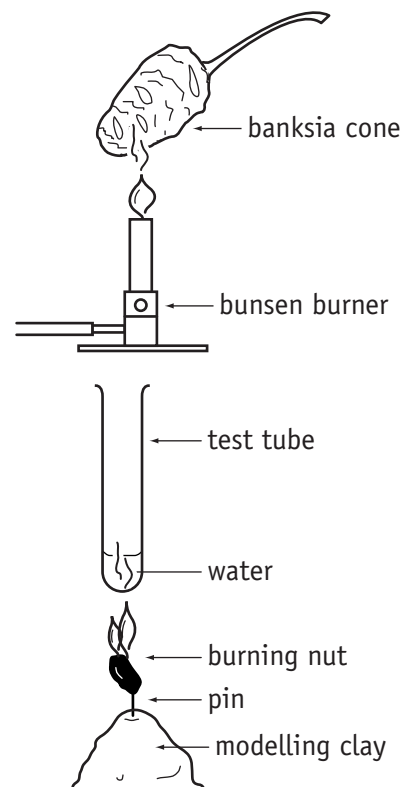
**| The kangaroo rat and the flour beetle obtain ALL their water from food such as seeds. They never drink.**

**| Some seeds can remain alive for many hundreds of years. Even wheat seeds found inside the Egyptian pyramids have sprouted when watered.**

**| Peach pits contain cyanide.**

## LINKED ACTIVITIES

- 1 **Simulate a bushfire.** *TEACHER DEMONSTRATION ONLY*  
*Safety glasses and laboratory coat should be worn.* Hold a mature banksia cone in a Bunsen flame for a few minutes. *(The seed pods will slowly open and the winged seeds can be removed. The seeds have a better chance of growing after a bushfire because competing plants have been destroyed.)*
- 2 **Compare the energy in two nuts.** *Safety glasses and laboratory coats should be worn.* Weigh a pecan nut, then hold it in position with a pin and light it with a match. Use the burning nut to heat a small amount of water in a test tube. Measure the change in temperature of the water. Repeat the above, using a peanut. *(Nuts are seeds that are extremely rich in proteins, minerals and vitamins; their oil content burns readily.)*



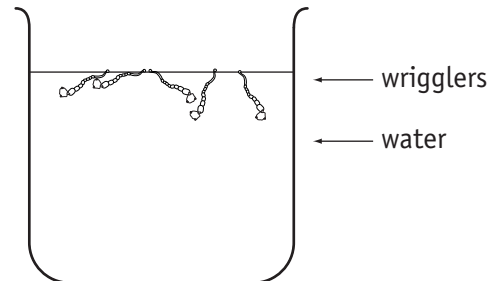
## 11

## MOSQUITO WRIGGLERS

You can make mosquito wrigglers dive on command.

## HOW TO DO IT

- 1 Collect some mosquito wrigglers from some stagnant water (look in ponds, drains, puddles, old buckets, old 44 gallon drums etc.) and put about six in a beaker of fairly clear water.
- 2 Leave the wrigglers undisturbed until they come to the surface and hang by their breathing tubes.
- 3 Tap the beaker with a pencil. (*The wrigglers will dive.*)



## EXPLANATION

Mosquito wrigglers respond to vibrations by diving, probably to avoid predators. However, after continued stimulation, they do not respond as readily.

## TO INVESTIGATE

What other stimuli will make wrigglers dive? (Try dipping a pencil into the water, a sharp clap, blowing on the water or passing a shadow over the beaker.) Also, how deep can wrigglers dive?

**If you get bitten by a mosquito, then you know it was a female. Male mosquitoes feed only on flower nectar.**

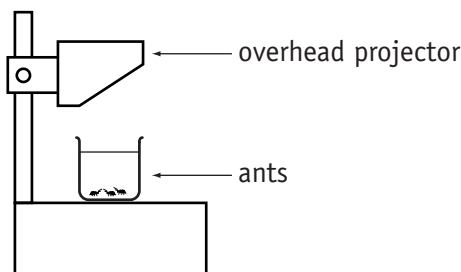
**Butterflies look fragile, but they have existed in their present form for millions of years longer than humans.**

## HISTORICAL NOTE

In 1900 Dr James Carroll suspected that mosquitoes carried the deadly disease yellow fever, so he bravely let an infected mosquito feed on him. He became seriously ill, so the experiment was a success.

## LINKED ACTIVITIES

- 1 **Resuscitate some ants.** Put some large ants in a jar, cool the jar until the ants are motionless, then place it on an overhead projector. (*All the chemical reactions in living things are controlled by enzymes, which decrease in activity as the temperature drops below their optimal level. As the ants are warmed by the projector they will become active again and can be observed on the screen.*)



- 2 **Observe predation.** Place a praying mantis in a jar with some small branches to climb on. Add a few small living insects. (*The mantis will readily strike at them.*)



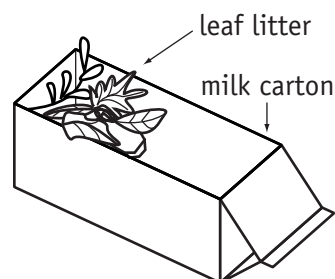
**During mating, the female praying mantis usually kills and eats the male.**

A handful of forest litter can conceal a surprising number of these creatures.

## HOW TO DO IT

### TEACHER DEMONSTRATION ONLY

- 1 Cut away one side of a milk carton to make a rectangular viewing tray. Put a handful of mixed leaf litter (bits of dry leaves and bark) in one half of the carton and leave the other half empty.
- 2 Collect 50 healthy woodlice (slaters) from leaf litter or under paving stones and logs in the garden. *Safety warning: wear sturdy gloves because poisonous spiders also inhabit these areas.*
- 3 Place all the woodlice on the litter and give them 2 minutes to hide.
- 4 Ask the students to observe the box and estimate the numbers of woodlice inside. *(They will usually underestimate the number.)*



## EXPLANATION

Woodlice avoid light and open spaces and are sensitive to slight differences in humidity. This tends to keep them in the leaf litter, where they are more protected from desiccation. They feed on decaying plant material.

## TO INVESTIGATE

What happens if the woodlice are placed in the bare half of the box, rather than in the leaf litter?

**The giant crab of Japanese waters measures over 3 m from limb tip to limb tip, and is a cousin of the woodlouse.**

**Funnel-web spiders can't kill dogs or cats. In fact, humans and other primates are the only vertebrates susceptible to funnel-web toxin.**

## LINKED ACTIVITY

Have an arthropod scavenger hunt. By scavenging in a suitable garden, each team of three should get as many points as they can from the list below. Collect only one of each item per group. *Safety warnings: do not collect spiders, bees, wasps or centipedes; redback spiders are common, so don't lift stones, logs or other objects with your bare hands.*

| List of Items     | Points | List of Items  | Points | List of Items | Points |
|-------------------|--------|----------------|--------|---------------|--------|
| praying mantis    | 100    | ant            | 20     | fly           | 30     |
| beetle (adult)    | 40     | beetle (larva) | 40     | caterpillar   | 40     |
| butterfly or moth | 40     | millipede      | 50     | slater        | 30     |
| aphid             | 50     | cockroach      | 40     | mosquito      | 50     |
| mosquito larva    | 60     | shield bug     | 50     | grasshopper   | 50     |

### HISTORICAL NOTE

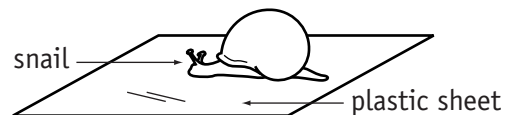
In the days of sail, crustaceans were used to torture people. A standard punishment on board British naval vessels was to use a rope to drag miscreants under the keel of the ship from one side to the other. In this process (known as 'keel hauling') the victims' skin would be lacerated by contact with the hard, sharp outer shells of the barnacles (crustaceans) that grew on the underside of the ships.

**CREATIVE LITERACY** Can you think of an arthropod for every letter of the alphabet?

You can watch a snail make waves.

### HOW TO DO IT

- 1 Put a snail in a dish with a little water to encourage it to come out of its shell.
- 2 Put the snail onto a sheet of damp clear plastic (e.g., from an overhead projector). As the snail is moving across the plastic sheet, look at it from the underside. (*A series of waves will be flowing forwards along its foot.*) *Safety warning: wash your hands after handling specimens.*



### EXPLANATION

Each time one of the ripples reaches a muscle it contracts and lifts a small part of the foot, which is then carried forward and set down. In this way the snail is propelled forwards.

### TO INVESTIGATE

Can a snail move along the edge of a safety blade without cutting itself?

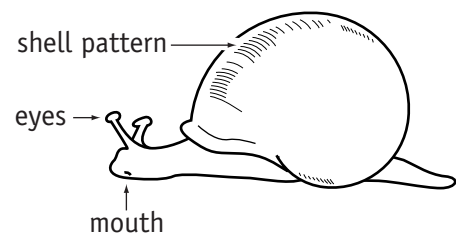
**You should be glad you're not a snail. A snail's anus is located just above and just behind its head.**

**Snails have both male and female sex organs. Before copulation one snail fires a hard calcium 'love dart' into the body of the other snail, then both snails donate sperm.**

**Our garden snail, *Helix aspersa*, is considered a delicacy in parts of Europe and the Middle East.**

### LINKED ACTIVITIES

- 1 **Look a snail in the eyes.** Observe what happens to a snail's eyes (which are located on the long tentacles) as it emerges from its shell. (*The tentacles extend without the eyes, then the eyes travel up the tentacles to the tips.*)
- 2 **Solve the following problem.** Does each snail have a unique shell pattern? How would you classify the types of patterns on their shells?
- 3 **Examine other molluscs.** Obtain specimens of mussel and oyster (in their shells), squid and octopus from a fish shop, and a slug from the garden. What features would you use to classify them into different groups of molluscs? (*Snails and slugs have a distinct head; mussels and oysters have a shell in two parts; and cephalopods have several tentacles.*)



**CREATIVE LITERACY** Create a crossword based on the theme 'molluscs'.

#### HISTORICAL NOTE

For hundreds of years there have been stories about sea serpents, such as the Loch Ness monster. Scientists believe that many of these may be due to sightings of giant squid (*Architeuthis*) which can be up to 20 m long.

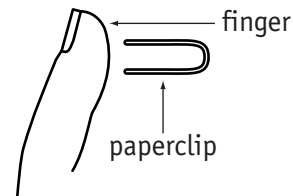
## 14

## SKIN PRICKS

The skin on the back of your hand cannot differentiate between one touch and two touches.

## HOW TO DO IT

- 1 Bend a paperclip into a U shape so the tips are about 6–7 mm apart.
- 2 Ask your partner to close his/her eyes and stretch out one hand, palm up. Then lightly touch the two tips onto your partner's fingers near the fingertips. *(Your partner will feel two distinct touches.)*
- 3 Turn your partner's hand over and explain that you will now touch the skin with either one or two tips. *(When you use two tips your partner will only feel one.)*



## EXPLANATION

Some parts of your skin have relatively few pressure receptors. In these areas, the spaces in between the receptors are insensitive to touch. The front of the hand, and especially the fingertips, have many more touch receptors than the back of the hand.

## TO INVESTIGATE

Which parts of your skin are least sensitive to touch? *(Your face and fingertips contain many touch receptors, but the upper arm, the small of the back and the back of the hand have far fewer.)*

**I** A normal person has more bacteria on their skin than there are people on Earth.

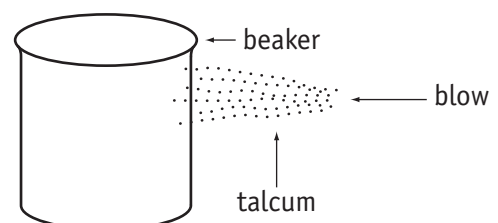
**I** Your perspiration contains some urea (although most of it is in your urine).

## LINKED ACTIVITIES

- 1 Classify the print of your index finger as an arch, loop or whorl. Prepare a bar graph to show the proportions of these in the class. *(The pattern of ridges on the skin helps to increase friction when you are holding on to things. The normal population is 65% loops, 30% whorls, and 5% arches. Every finger of every person, including identical twins, has a unique print.)*



- 2 **Identify the culprit.** Have an unidentified staff member put a clear fingerprint on a glass beaker. Lightly puff talcum powder onto the glass, so the fingerprint becomes visible. Stick some clear adhesive tape over the print, then peel the tape off. Hold the piece of tape up to the light, so the pattern of the print is visible. Identify the staff member from a full set of inked prints of all the staff. *(The pattern of a fingerprint is left on a glass because the pores of the skin normally exude small amounts of fats and oils, which are deposited on the glass in the shape of the ridges.)*



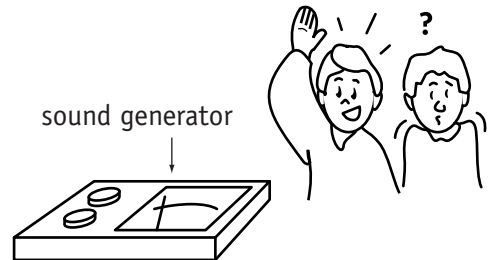
## 15

## HEARING RANGE

There are sounds you cannot hear.

## HOW TO DO IT

- 1 Use a sound frequency generator to find a sound all the students in the class can hear comfortably.
- 2 Tell all the students to put their hands in the air and only lower them when they cannot hear the sound any more. Increase the frequency until most, but not all, of the students cannot hear it.
- 3 Demonstrate that it is possible to increase the frequency to much higher levels that no-one can hear.



## EXPLANATION

Humans can only hear sounds up to about 20,000 Hz. Most of the sounds grasshoppers, dolphins and bats make are well above 20,000 Hz. As you get older, the range of sounds you can hear decreases.

**Dogs and cats can hear many sounds we cannot.**

## HISTORICAL NOTE

Beethoven was profoundly deaf during most of his composing years.

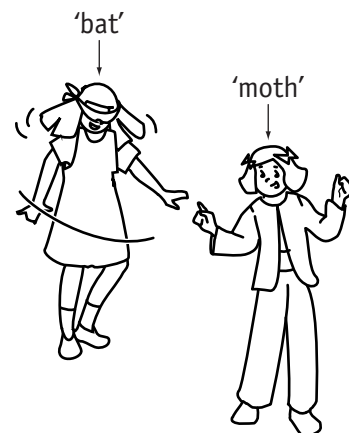
**CREATIVE LITERACY** Write a patent application for this invention that is intended to silence a neighbour's barking dog. The device is attached to the outside of your house and is activated by the sound of a dog barking. Then it emits a piercing sound that is annoying to the dog, but inaudible to humans.

## LINKED ACTIVITIES

- 1 **Enhance your hearing.** Stand 2–3 m away from a ticking clock. Face the clock and cup your hands behind your ears. *(The ticking will appear to be slightly louder. The outer ear of humans plays little or no role in helping us hear, but ear trumpets and other sound collection devices can effectively increase audibility.)*



- 2 **Dramatise echolocation by bats.** *This activity should be done outside in a flat open area free of obstacles. Mark out an open space about 3 m square. Nominate one student to be the bat and another to be the moth, which tries to escape. The bat is blindfolded. Every time the bat says "bat" the moth must say "moth" (so if the bat says 'Bat-bat-bat-bat...' in rapid succession the moth must reply every time). (Even though he/she cannot see, the bat will find it surprisingly easy to catch the moth, and thereby demonstrate the effectiveness of this hunting technique.)*



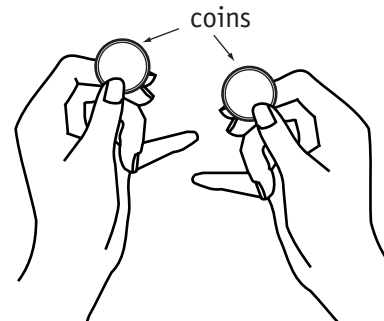
## 16

## SEEING THINGS

Your eyes can double your money.

## HOW TO DO IT

- 1 Hold two \$2 coins about the length of this page from your eyes, one coin in each hand.
- 2 Look past them to the wall of the room. (*You will see four coins.*)



## EXPLANATION

When we look at an object, each eye sees it from a slightly different angle and this gives us stereoscopic vision. Our brain interprets the information the eyes send to it but can sometimes make mistakes, especially when depth perception is involved (as in the activity above).

**| Many spiders have eight eyes.**

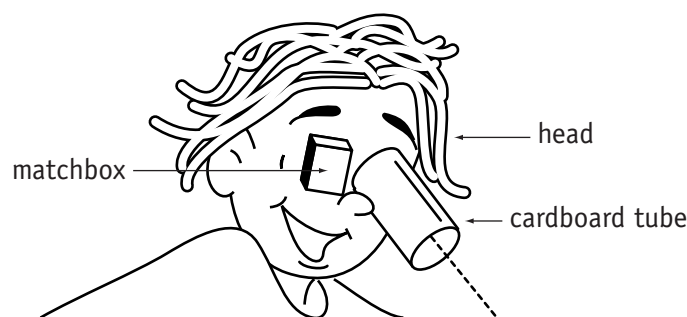
**| The closest star to earth cannot be seen with the naked eye. Proxima Centauri is a relatively dim star that is only slightly bigger than the planet Jupiter.**

**| The furthest a person can see is about 2 million light years. This is the distance to Andromeda, a neighbouring galaxy, which is visible on a clear night.**

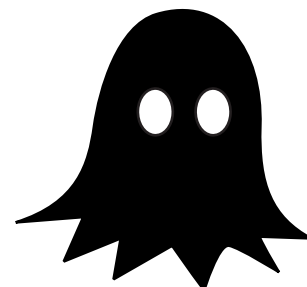
**| Blindness or partial blindness affects 28 million people worldwide.**

## LINKED ACTIVITIES

- 1 **See through a matchbox.** Collect a cardboard tube from a toilet roll, put it to one eye and read this sentence through it with both eyes open. Now hold a matchbox in front of your other eye. (*You will seem to be reading through a hole in the matchbox.*)



- 2 **Spot a spook.** Stare intently at this picture for about 1 minute without moving your eyes. Then look up at the bare ceiling and stare fixedly at one part of it. (*You will see an 'after-image' of the ghost. Overstimulation by staring at the picture causes the retina to become fatigued and unable to adjust quickly when your gaze is shifted.*)



## 17

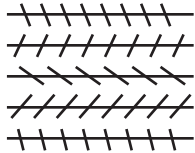
## OPTICAL ILLUSIONS

Some patterns of lines can fool your brain.

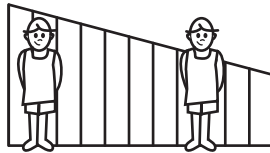
## HOW TO DO IT

Answer the questions below, first by using a sight estimation and then by measuring with a ruler.

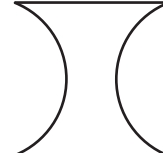
Are these lines parallel?



Is one person taller?



Is this vase taller than it is wide?



## EXPLANATION

The brain automatically makes corrections to the images it receives from the eye and this helps us to interpret our surroundings. (For example, as objects get further away they look smaller, but we understand that they are still the same size.) But when unusual images are presented to the brain, it can give us the wrong information.

**A young baby's head makes up about one-quarter of the length of its body. In some parts of Africa, a child is deemed to be old enough to begin school when he/she can put one arm over his/her head and touch the other shoulder. Younger children's heads are too big to allow them to do this.**

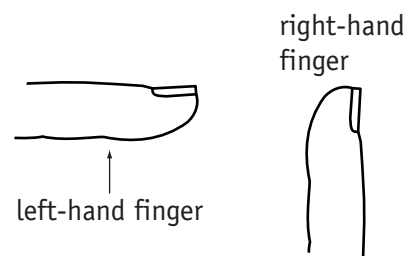
## HISTORICAL NOTE

Real human size has been steadily increasing this century. For example, British people are, on average, 11 cm taller than last century. However, the increase in Europeans and North Americans has been twice as much as among Australians. Nobody knows why.

## LINKED ACTIVITIES

**1 Use only one eye to estimate distance.** Close your right eye and hold up a finger from your right hand, about half a metre in front of your body. Position your left hand about the same distance away, but much further to the left. Slowly move a finger from your left hand towards your right hand finger and try to touch it. *(At the first attempt you will probably miss because we normally use both eyes to estimate distance.)*

**2 Measure your body.** Use a length of cotton to measure the circumference of your wrist. Compare this with the circumference of your neck. *(The length will be roughly twice as long.)* Is there another body part that is twice as long again?



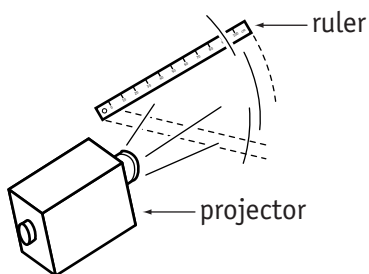
**Astronauts get taller when they are in space. This is because there is no effect of gravity, so the discs between their vertebrae are not compressed as much.**

## MAGIC PICTURE

You can create an image in mid-air.

## HOW TO DO IT

- 1 Insert a slide into a projector and adjust the lens so that the picture is focused at about 1 m in front of the projector. Take the screen away.
- 2 Hold a ruler 1 m in front of the projector and move it quickly from side to side. (*You will see the full picture where you are waving the ruler.*)



## EXPLANATION

At each position of the ruler, a part of the picture can be seen on it. When you wave the ruler these images succeed each other rapidly and your brain 'sees' the current image as well as the one it already held from a fraction of a second before. It puts them together to create the illusion of seeing the whole picture.

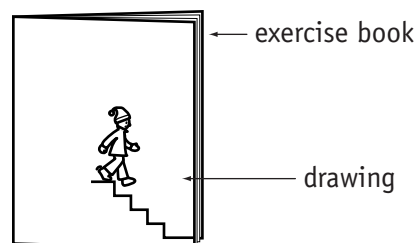
**IA high speed camera can photograph a bullet in flight.**

## HISTORICAL NOTE

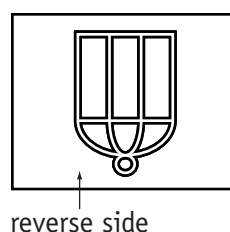
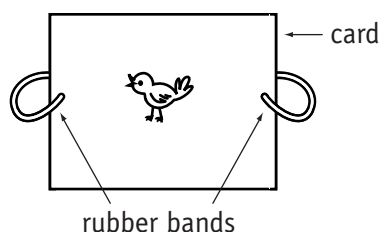
Old movies used to be known as 'the flicks' because the images were presented at a slower speed that was not quite fast enough to be perceived as a smooth motion. A modern motion picture is simply a sequence of photographs presented in rapid succession, at 24 frames per second.

## LINKED ACTIVITIES

- 1 **Make a cartoon** by drawing a sequence of pictures, each one slightly different to the last, on the pages of an exercise book. Then flip the pages. (*When the pictures are presented in rapid succession it will create the illusion of smooth motion.*)
- 2 **Put the bird in the cage.** Cut out a piece of plain cardboard about the size of a playing card. Draw the bird on one side and the cage on the other. Tie a rubber band to each end of the card, wind it up tightly, then let it unwind. (*As the card rapidly unwinds, the bird will appear to be in the cage.*)



**CREATIVE LITERACY** Debate the issue 'Cartoons and motion pictures have had a positive effect on society'.



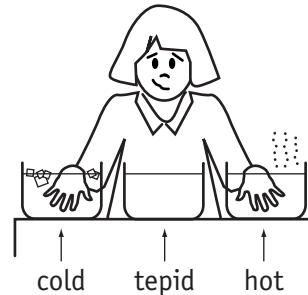
## 19

## BOWLS OF WATER

One hand feels warm and the other hand feels cold, but they are both in the same bowl of water.

## HOW TO DO IT

- 1 Set up three empty ice cream containers in a row. Put hot (but not scalding) water in the first, tepid water in the second and cold water in the third.
- 2 Put your left hand in the hot water and your right hand in the cold water for about 1 minute. Swirl the water a little with your hands to keep the temperature even.
- 3 Quickly transfer both hands into the tepid water and concentrate on what each hand is sensing. (*The left hand will sense cool water, and the right hand will sense warm water.*)



## EXPLANATION

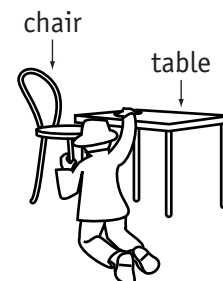
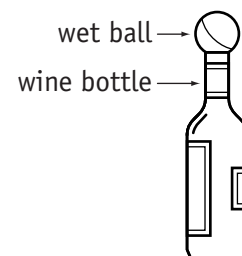
Your heat and cold receptors sense changes in temperature rather than absolute temperature. The tepid water feels cold by comparison with hot water.

**Most animals don't perspire.** In fact, humans and horses are among the very few mammals that do use perspiration to cool themselves. Instead, dogs pant to evaporate water from their tongues (dogs pant when they are hot rather than 'puffed'), and kangaroos lick the underside of their wrists where the blood vessels are near the surface.

**Rattlesnakes can locate warm-blooded prey by sensing the heat they give off.** Their heat sensors can detect a temperature increase of 0.002°C.

## LINKED ACTIVITIES

- 1 **Use body heat to expand air.** Place an empty wine bottle in the fridge until it is cold. Take it out and place a wet ping-pong ball in the mouth of the bottle. Wrap both of your hands around the bottle. (*The air inside the bottle will expand as it is warmed by your body heat. Some of the air will leave the bottle, causing the ball to plop up and down in the opening.*)
- 2 **Feel heat conduction.** Use your hand to compare the temperature of a metal chair leg and a wooden table top. (*The metal will feel colder because the heat from your hand is rapidly conducted away. The wood will feel warmer because it is a poor conductor, so the heat from your hand is not dissipated.*)



**CREATIVE LITERACY** Imagine that you are an early explorer in the Antarctic. Write a diary entry explaining the steps you are taking to maintain your body heat.

# BODY BALANCE

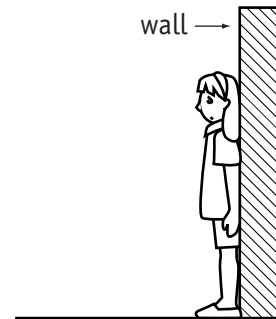
When you have your heels against the wall you cannot touch the floor.

## HOW TO DO IT

- 1 Stand with the backs of your heels firmly against the wall.
- 2 Try to touch the floor in front of your feet without moving your feet or bending your knees. *(You will not be able to touch the floor without falling over.)*

## EXPLANATION

A person's centre of gravity is usually just above his/her hips. When you try to touch the floor your centre of gravity moves in front of both feet and you topple over.



**Tug-of-war teams try to achieve a low centre of gravity. The five-man tug-of-war was an Olympic event up until 1920.**

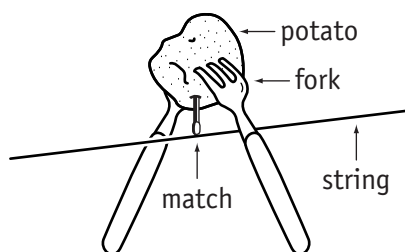
**CREATIVE LITERACY** Sit in a chair with your feet on the ground. Try to stand up without leaning forward or using your arms. Write a simple explanation for what you have found.

## HISTORICAL NOTE

Late last century, a Frenchman named Charles Blondin became famous for performing above Niagara Falls on a tightrope over 300 m long and 50 m high. He made the crossing several times with variations that included being blindfolded, walking on stilts, pushing a wheelbarrow and piggybacking his agent.

## LINKED ACTIVITIES

- 1 **Make a tightrope walker.** Insert two forks and a match into a potato so it balances on a tight string. *(Using the weight of the fork handles, the centre of gravity of the system will be lowered to just below the string, so the potato will not fall off.)*



- 2 **Which senses are involved in balance?** Close your eyes then try to balance on one leg. *(You will find it extremely difficult to stay upright. The eyes and the semicircular canals of the middle ear work together to give you information about balance.)*



**Your brain makes you seasick.** When your eyes and your middle ear send conflicting messages (as when you are in a rolling ship), the brain gets confused and floods your body with stress hormones that make you feel sick. This seems a pointless thing to do and scientists still don't understand why the brain does it.

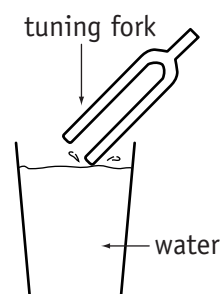
## 21

## TUNING FORK

A tuning fork will make water splash up.

## HOW TO DO IT

- 1 Strike a tuning fork firmly.
- 2 While it is still ringing strongly, dip the tips of the prongs into some water in a plastic cup. Don't let the prongs touch the sides of the cup. (*The water will splash up.*)



## EXPLANATION

The sound of the tuning fork is made by vibrations of the prongs. The water splashes when energy from the vibrations is transferred to the water.

## TO INVESTIGATE

What do the vibrations of a tuning fork do to a ping-pong ball? How would you use an overhead projector to observe the vibrations of a tuning fork?

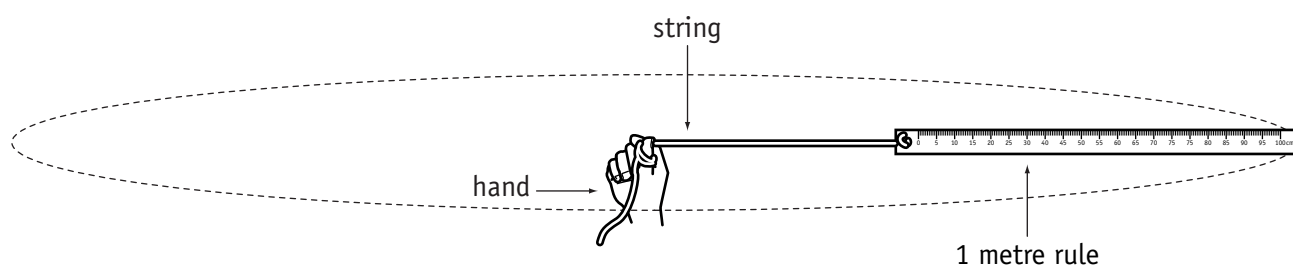
**CREATIVE LITERACY** Imagine that your house is under the flight path of a major airport. Write a letter to inform your council about the problem of vibrations.

**Giraffes do have a voice, but it is used so rarely that for many years scientists assumed them to be completely incapable of vocalising.**

**A stockwhip makes a sonic boom. When a stockwhip cracks, the tip of the whip is travelling faster than the speed of sound, and there is a crack as it breaks the sound barrier.**

## LINKED ACTIVITIES

- 1 **Make a cellophane whistle.** Cut a piece of cellophane about 3 cm square, making sure that one edge is straight. Pull the straight edge taut between your hands, hold it to your mouth and blow firmly onto the edge. (*The blowing air makes the edge of the cellophane vibrate with a high frequency, making a high-pitched whistle.*)
- 2 **Make a bullroarer.** *TEACHER DEMONSTRATION ONLY* — Make a hole at one end of 1 m rule. Cut 1 metre of strong string and tie one end to the hole (securely!). Hold the other end of the string and whirl it around your head. *Safety warning: this activity should be done outside the classroom in a location where students can observe without risking injury if the string breaks. (The rule will vibrate as it moves through the air, producing a loud whirring sound.)*



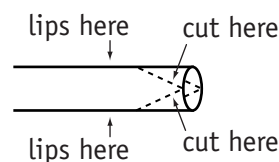
## 22

## STRAW CLARINET

You can make sounds with a straw.

## HOW TO DO IT

- 1 Flatten the end of a soft plastic straw and cut the two corners off so there are two points at the end.
- 2 Close your lips firmly over the end of the straw, so the points are just inside your mouth, and blow into the straw.  
*Safety warning: sit down and hold the straw firmly to prevent it being pushed too far into your mouth by accident. (With a bit of practice you will make a loud, buzzing sound.)*



## EXPLANATION

When you blow, the points vibrate like the reeds in a clarinet. This makes the column of air inside the straw vibrate. The pitch of the sound depends on the length of the column of air.

## TO INVESTIGATE

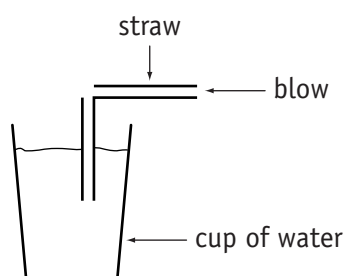
What would happen if you used a pair of scissors to cut the straw in half while you were still blowing?

**|To play the didgeridoo properly, you need to breathe in and blow out at the same time.**

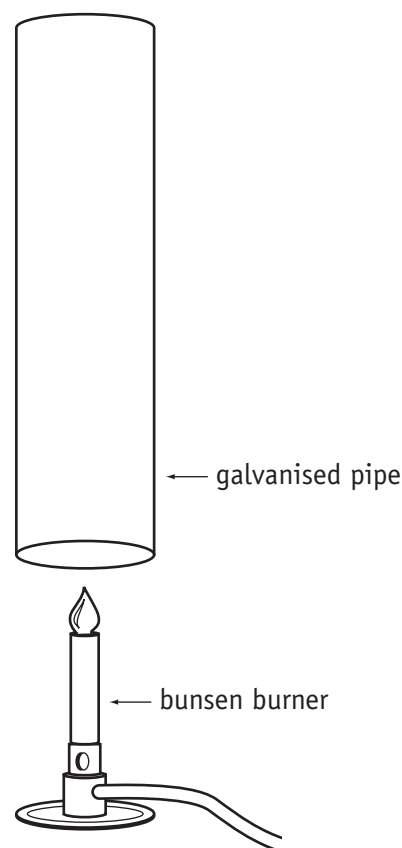
**|The column of air in a French horn can be over 5 m long.**

## LINKED ACTIVITIES

- 1 **Make a bird whistle.** Make a cut in a straw and bend it as shown in the diagram below. As you puff intermittently through the horizontal section, move the vertical section up and down in the water. *(The sound will become more high-pitched when there is a shorter column of air vibrating inside the vertical section.)*



- 2 **Amplify a Bunsen burner.** *TEACHER DEMONSTRATION ONLY*  
*Laboratory coat and heatproof gloves should be worn. Light a Bunsen burner and carefully lower a galvanised pipe over it. (The sound of the Bunsen burner will resonate inside the pipe.)*

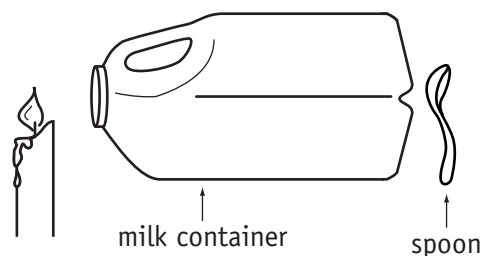


## BLOW OUT A CANDLE

A sound wave can blow out a candle.

## HOW TO DO IT

- 1 Obtain an empty, plastic 2 L milk container and remove the lid.
- 2 Position the opening of the container about 2 cm to one side of a candle flame.
- 3 Tap the base of the container with a teaspoon. *(The flame will be extinguished.)*



## EXPLANATION

The tap on the end of the container produces a compression wave that travels through the container and is funnelled through the opening with enough force to blow out the candle.

## TO INVESTIGATE

Cut a hole in the base of a large, plastic garbage bin. Put the lid on the bin and tap it with your hand. From how far away can it knock over a polystyrene cup?

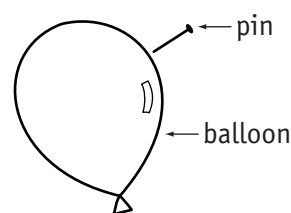
**Loud sounds can momentarily stun people. This fact is used by karate experts, who are trained to shout loudly as they fight.**

## HISTORICAL NOTE

Possibly the loudest sound ever made was the volcanic explosion on Krakatoa, an island in Indonesia, in 1883. The sound of the explosion was heard over 3,200 km away in Australia. The eruption also caused a tsunami (tidal wave) about 40 m high which killed 36,000 people on neighbouring islands.

## LINKED ACTIVITIES

- 1 **Use a balloon to model thunder.** *Safety warning: give others fair warning of the noise and don't hold it near anyone's ear. Blow up a balloon until it is half inflated, then burst it with a pin. (The air is slightly compressed inside the balloon, so there will be a bang when it expands rapidly. This is similar to the way thunder is produced — a bolt of lightning heats the air around it, which suddenly expands, and we hear this compression wave as thunder.)*



**CREATIVE LITERACY** Explain what causes the noise when you blow up a paper bag then 'bang' it with your hand or when you stamp on an empty flavoured milk carton to make it bang.

- 2 **Calculate how far away a thunderstorm is.** In a thunderstorm, lightning and thunder are produced at the same time. We see the lightning instantaneously, but it takes time for the sound of thunder to travel to us. *(Sound travels at about  $\frac{1}{3}$  km/s, so count the number of seconds between seeing the lightning and hearing the thunder, then divide by 3 to give the distance in kilometres.)*

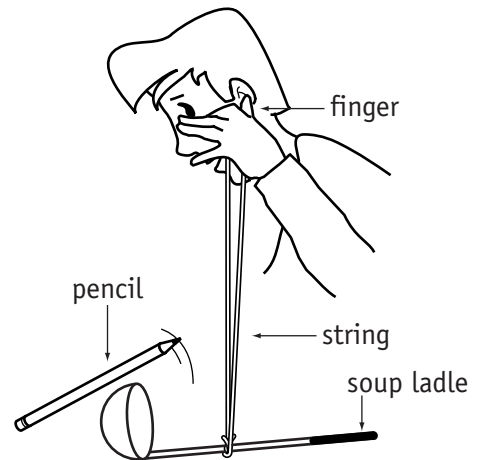
## 24

## A KITCHEN GONG

You can make a soup ladle sound like a gong.

## HOW TO DO IT

- 1 Tie the handle of a metal soup ladle to the middle of a 1 m length of string.
- 2 Hold the two free ends of the string firmly to the openings of your ears, letting the ladle hang between them.
- 3 Ask your partner to tap the cup of the ladle with a pencil. *(You will hear a deep ringing sound like a gong, but your partner will not.)*



## EXPLANATION

The ladle has a rounded metallic shape like a gong or a bell. You hear the sound loudly because the sound travels more quickly through the solid string than through the air.

## TO INVESTIGATE

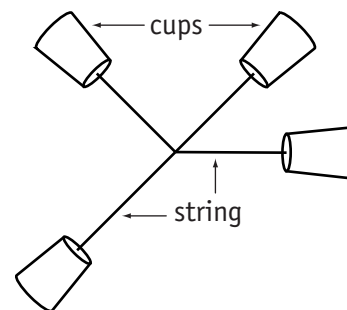
Would the sound be different if you used a teaspoon or a tablespoon instead of the ladle? What would two spoons together sound like?

**Sound travels 15 times faster through solid steel than through the air. This is why noises in the school central heating pipes are often clearly heard in other parts of the school.**

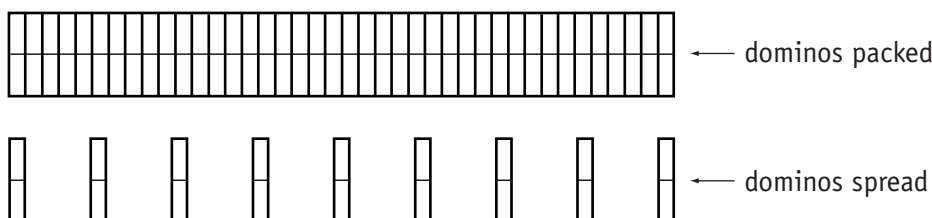
**Some low-pitched sounds made by whales and dolphins can travel over 100 km under the water.**

## LINKED ACTIVITIES

- 1 **Make a string telephone** by connecting two polystyrene cups with a length of string. Compare the sound transmission over 10m of string with that over 2m. Compare the sound transmission in a multi-channel link-up with four cups connected together. What happens if the string goes around a corner?
- 2 **Set up a domino model of sound travel.** Make two lines of dominoes standing on edge. In one line, the dominoes should be packed together so they are touching each other (to represent a solid) but in the other line they should be spaced apart (to represent a gas). The lines should be exactly the same length. Tip the first domino in each line. *(The 'chain reaction' will be faster where the dominoes are packed together.)*



**CREATIVE LITERACY** On the moon there is no air to carry sound. Write a description of how two astronauts could talk to each other on the moon, other than by radio.



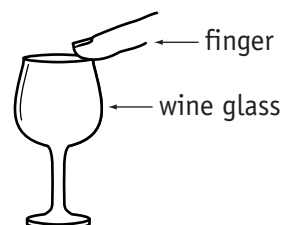
## 25

## RUB A WINEGLASS

You can make a musical note by rubbing a wineglass.

## HOW TO DO IT

- 1 Put a crystal wineglass on the table and hold it by the neck rather than the base.
- 2 Wet a finger and rub it smoothly in one direction around the rim of the wineglass. *(With a bit of practice a musical note will be heard.)*



## EXPLANATION

The rubbing causes vibrations in the glass. When these vibrations match the natural frequency of vibration of the wineglass, the glass will vibrate at that frequency (that is, resonate).

**Sound cannot travel through a vacuum, so outer space is always silent.**

**Some people believe that plants respond to music and there have even been PhD theses written about it. The results of one study showed that many flowers prefer classical music and jazz, definitely do not like loud rock music and are unaffected by country and western.**

## HISTORICAL NOTE

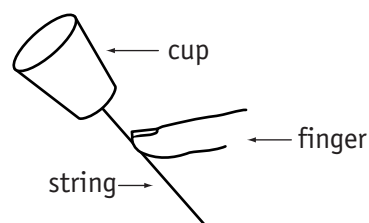
Resonance can destroy bridges. In 1940 the Tacoma Narrows Bridge, a suspension bridge about 1 km long, began vibrating in response to a steady wind blowing across it. The vibrations built up until the bridge tore itself apart and collapsed into the water.

## LINKED ACTIVITIES

- 1 **Rattle a needle.** Obtain two wineglasses that both make exactly the same note. (You may need to add water to one to change its note.) Stand a sewing needle in one of the glasses. Wet your finger and rub the rim of the other glass, which should be very close to but not touching the first glass. *(The first glass will resonate and its vibrations will make the needle rattle against the glass.)*



- 2 **Use resonance to amplify sounds.** Tie a length of string to the base of a polystyrene cup. Pull the string taut to the cup and stroke it with the skin of your fingertip. Then let the string go slack and stroke it. *(The sound will only be audible when the vibrations are transferred to the cup by taut string. The sound waves reflect and rejoin inside the cup to become amplified.)*



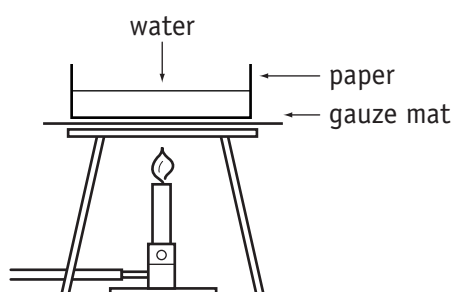
## PAPER BEAKER

You can heat water in a paper container.

## HOW TO DO IT

*TEACHER DEMONSTRATION ONLY Safety glasses and laboratory coat should be worn.*

- 1 Use photocopier paper to make a paper tray with a base of about 10 cm square and sides about 3–4 cm high. Try to avoid puncturing the paper.
- 2 Half fill it with water, then place it on a gauze mat and tripod.
- 3 Heat it with a Bunsen. (*The paper will not burn.*) *Safety warning: the paper may disintegrate as the water approaches boiling, so it is important to turn the Bunsen off well before this point.*



## EXPLANATION

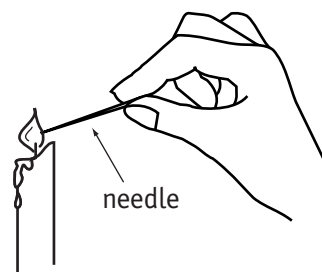
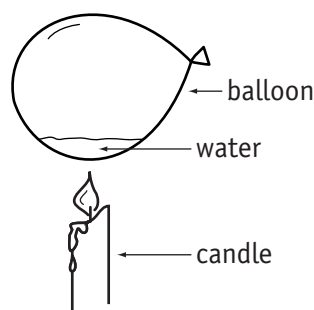
Most of the heat is conducted to the water, so the paper does not reach its ignition temperature.

**| Diamond is the best thermal conductor. Silver is a better thermal conductor than copper.**

**| The heat in the interior of the earth is mainly produced by radioactivity.**

## LINKED ACTIVITIES

- 1 **Heat water in a balloon.** Pour some water into an empty balloon, then blow it up with the water inside and tie the opening. Lower the base of the balloon onto a candle flame. (*The heat is conducted to the water, so the balloon will not burst.*)
- 2 **Feel heat conduction.** Hold a sewing needle at the blunt end, and position its point in a candle flame. (*After a few moments you will feel the heat which has been conducted through the needle.*) *Safety warning: the needle will quickly become too hot to hold, so be ready to place it on a heat mat before it burns you.*

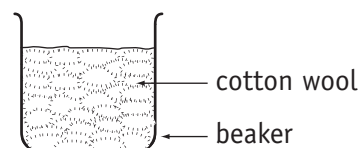


**CREATIVE LITERACY** Write an 'Ode to a Saucepan' explaining the good conductors and poor conductors in its design.

You can squash a surprising amount of cotton wool into a beaker of water.

### HOW TO DO IT

- 1 Tightly pack cotton wool into a small beaker until it reaches the 100 mL mark.
- 2 Remove the cotton wool from the beaker and replace it with exactly 100 mL of water.
- 3 Slowly add all the cotton wool to the beaker, pushing it down firmly with a pencil to remove all the air. Make sure that none of the cotton wool is above the surface. (*The water level will only rise 3–4 mL.*)



### EXPLANATION

Materials that contain a lot of air are usually good insulators. Cotton wool consists mainly of air, and the amount of cotton is relatively small. When cotton wool is submerged, the water replaces the air, and the slight rise in the water level represents the solid portion.

**Emperor penguins have to stand on the bare ice throughout the long Antarctic winter, yet they have absolutely no insulation on their feet.**

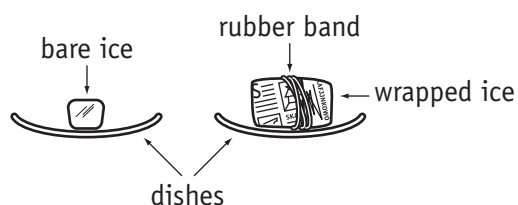
### HISTORICAL NOTE

Asbestos causes mesothelioma, a fatal form of lung cancer. Yet asbestos used to be used in house insulation, school laboratory heat mats, fire-resistant clothing and even theatre curtains. It is a very poor conductor of heat.

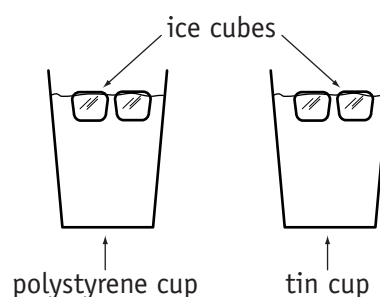
**CREATIVE LITERACY** How does fur keep animals warm? Write your arguments as to why people should or should not wear fur coats.

### LINKED ACTIVITIES

- 1 **Insulate an ice block.** Wrap an ice block in a large sheet of newspaper and secure it with a rubber band. Place another ice block on a dish, uninsulated. Compare the rates of melting. (*Air trapped in the newspaper will insulate the first piece of ice from the heat in the room.*)



- 2 **Investigate insulation in cups.** Obtain a polystyrene cup and a tin cup of about the same size. Fill both with water and put two ice cubes in each. Compare the rates of melting of the ice. (*The polystyrene contains air, which insulates the contents, so the ice will melt more slowly.*)



**CREATIVE LITERACY** It has been said that there is more heat in an iceberg than a candle flame. Discuss whether or not this could be true and how it could be verified.

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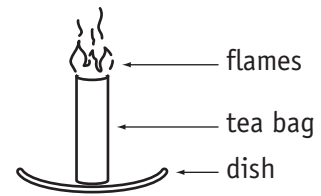
## HOT AIR TEABAG

A burning teabag will float into the air.

## HOW TO DO IT

**TEACHER DEMONSTRATION ONLY** Safety glasses and laboratory coats should be worn.

- 1 Remove the string and the staple from a teabag, empty the tea from it and unfold it to form a cylinder about 10 cm long.
- 2 Stand the cylinder upright on a dish and use a match to light it all the way around the top rim, so it burns evenly all the way down. If the teabag starts to topple over then use the spent match to lightly support it at the side. *(As the burning proceeds down towards the dish the teabag will suddenly lift 1–2 m into the air.)*



## EXPLANATION

The heat from the burning teabag causes the air above it to rise. When most of the material of the teabag has burned away it becomes light enough to be lifted into this convection current that has formed above it.

## HISTORICAL NOTE

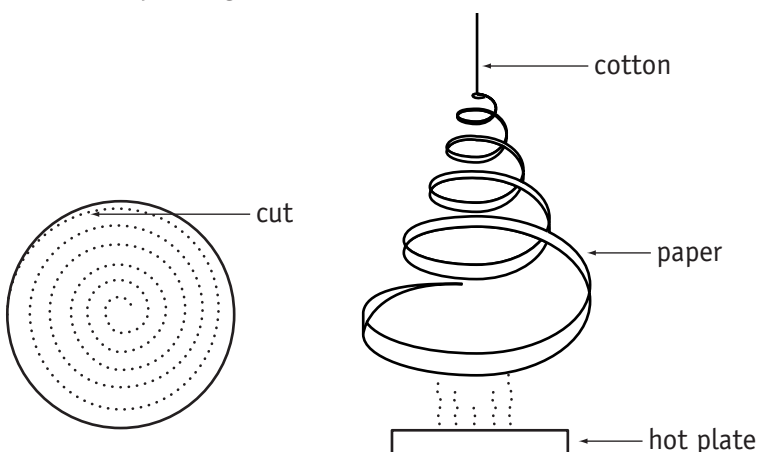
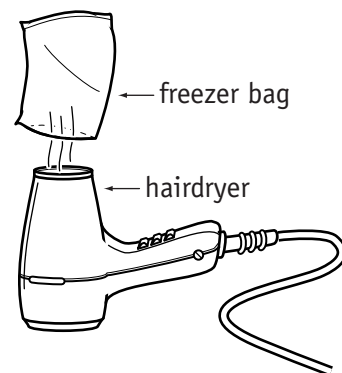
Early hot air balloons used straw and wool as fuel.

**It is difficult for aeroplanes to take off from high altitude airports on hot days. The reduction in air pressure, caused by both the high altitude and the expansion of air due to heat from the runway, means that some aeroplanes can only take off if they are half empty.**

**Dick Smith crossed Australia in a hot air balloon. He has been referred to as 'Australia's answer to Leonardo da Vinci' because he was also the first person to circumnavigate the world in a helicopter, and to reach the North Pole by helicopter. He hasn't owned Dick Smith Electronics since 1982.**

## LINKED ACTIVITIES

- 1 **Model a hot air balloon.** Obtain a kitchen freezer bag and use a hairdryer to fill it with hot air. Turn the hairdryer off then release the bag. *(The bag will rise above your head because the warm air inside is less dense than the surrounding air.)*
- 2 **Make a twister.** Cut a circle of paper about 15 cm in diameter, then cut into it turning it into a spiral. Tie a length of cotton to the centre and let it hang a few centimetres above a hot plate. *(The convection currents will start it spinning.)*



**CREATIVE LITERACY** Imagine that you are a wedge-tailed eagle. Describe your use of convection currents to gain altitude and remain aloft.

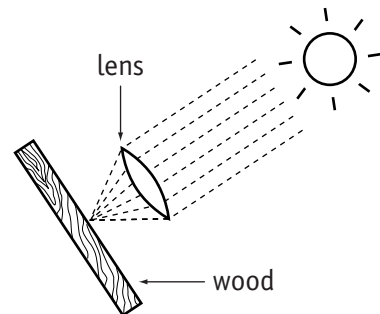
## HEAT RAYS

You can use a magnifying lens to burn wood.

## HOW TO DO IT

*Safety warning: never look directly at the sun, as this may cause permanent eye damage.*

- 1 Hold a convex lens in direct sunlight, so the rays are focused onto a dark piece of wood.
- 2 Move the lens up and down until the spot of light is as small as possible, then move the lens a few millimetres further away from the wood. *(The wood will turn black and begin to smoke.)*



## EXPLANATION

Light rays and infra-red heat rays travel from the sun. As they enter the convex lens they become refracted so they begin to converge. The focal point is where the rays meet. The focal point for heat is slightly further from the lens than the focal point for light.

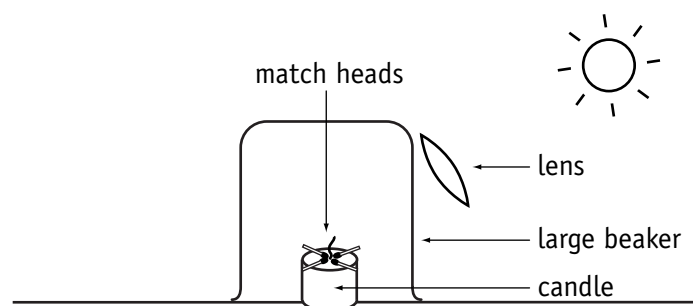
**The heat and light in this activity were produced in the centre of the sun and took over 100,000 years to travel to its surface, then about 8 minutes to travel the 150 million kilometres to earth.**

**The centre of the earth is hotter than the surface of the sun.**

**Just before the sun dies it will expand until its surface almost reaches the earth. This will destroy all life on our planet. Don't worry though, because it won't happen for another 5 billion years.**

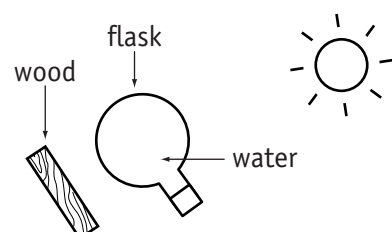
## LINKED ACTIVITIES

- 1 **Light a candle.** *TEACHER DEMONSTRATION ONLY* — Place an unlit candle upright on the ground and put four match heads next to the wick. Cover the candle with a large beaker. Use a convex lens to focus direct sunlight onto the match heads. *(The match heads will ignite, and so will the candle.)*



**CREATIVE LITERACY** Write a description of this demonstration. Use as much scientific jargon as you possibly can.

- 2 **Use water to burn wood.** *TEACHER DEMONSTRATION ONLY* Fill a round-bottomed flask with water and seal it. Use it to focus direct sunlight onto a dark piece of wood. *(The heat will be focussed onto the wood by the water in the flask.)*

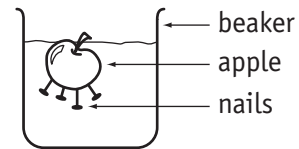


# FLOATING APPLE

This apple will sink in hot water but float in cold water.

## HOW TO DO IT

- 1 Obtain a fresh apple and some large nails. Embed enough nails in the underside of the apple so that it will only just float in cold water.
- 2 Place the apple in a large beaker of cold water. *(The apple will float.)*
- 3 Place the apple in a large beaker of hot water. *(The apple will sink.)*
- 4 Add cold water to the beaker with the sunken apple. *(The apple will rise to the surface.)*



## EXPLANATION

The hot water is less dense than the cold water (because water expands when it is heated), so the hot water has less upthrust. (Upthrust is reduced in fluids of lower density.)

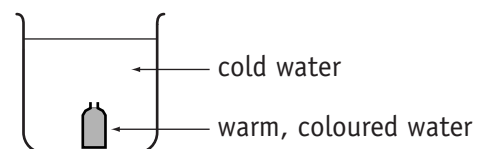
Heat causes the sea to expand. The El Niño effect is caused by a large pool of relatively warm water (5–10°C higher than normal) located in the eastern Pacific. Because this water is warm, it has expanded, so sea levels along the coast of South America can be up to 25 cm higher than usual.

**CREATIVE LITERACY** Research El Niño on the internet. Write a factual limerick about El Niño.

Some types of bacteria live in boiling water in hot springs.

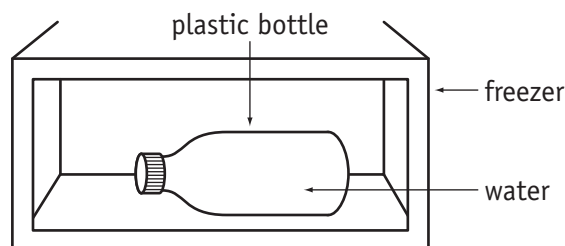
## LINKED ACTIVITIES

- 1 **Make an underwater convection current.** Take a small vial, add a few drops of food colouring and fill it to the brim with hot water. Place it in a large beaker of cold water. *(The coloured water is warmer and less dense, so it will move to the top of the beaker.)*



Underwater volcanoes make the continents move. Australia will have moved over 4 m northwards during your lifetime.

- 2 **Expand a bottle.** Fill a plastic bottle with water and screw the lid on tightly. Place it in the freezer overnight. *(Water expands as it freezes, so the bottle will be swollen or cracked in the morning.)*



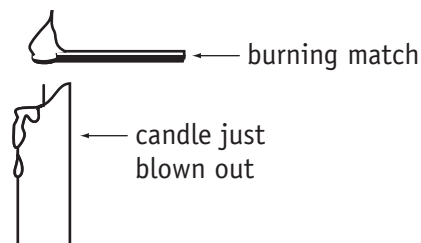
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## PARAFFIN GAS

You can light a candle without touching the wick.

## HOW TO DO IT

- 1 Secure a candle in a stable position and light it.
- 2 Strike another match and have it ready.
- 3 Blow the candle out and immediately hold the burning match 1–2 cm directly above the top of the wick. *(The wick will relight.)*



## EXPLANATION

Candle wax exists as solid, liquid and gas. Immediately after you have blown out the candle there is still some paraffin gas near the wick. This gas ignites when a flame is brought near it.

## TO INVESTIGATE

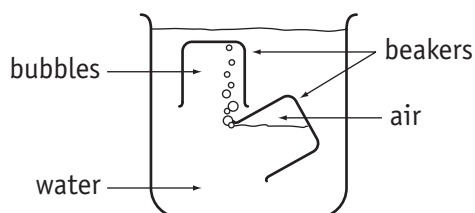
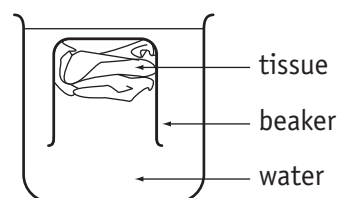
How does the appearance of paraffin change as it turns from a liquid to a solid? *(It changes from clear to white.)*

**Liquid petrol doesn't burn — the vapour does. But don't try to test this because the vapour will always be there. Safety warning: petrol is extremely dangerous and should never be near a naked flame.**

**Liquid water is common on earth but is one of the rarest substances in the universe.**

## LINKED ACTIVITIES

- 1 **Keep tissues dry under water.** Crumple a tissue and place it inside a beaker. Invert the beaker and place it in water. *(The tissue will remain dry because air prevents water from entering the beaker.)*
- 2 **Pour a beaker of air.** Add water to a very large beaker until it is almost full. Hold two small beakers inverted under the water — the first containing air and the second containing water. Hold the first beaker directly below the second and slowly tilt it. *(The gas will be transferred to the second beaker.)*



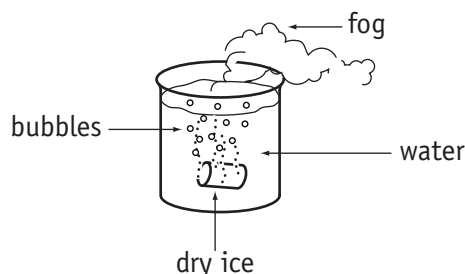
**CREATIVE LITERACY** Have a class discussion about the usage of the term 'gas' in everyday life. Do scientists use this term differently?

You can make fog come out of a beaker of water.

## HOW TO DO IT

### TEACHER DEMONSTRATION ONLY

- 1 Half fill a very large beaker with water.
- 2 Add about a dessertspoonful of dry ice. *Safety warning: do not touch the dry ice with your bare hands. (A dense white fog will emerge from the beaker and spread across the floor.)*



## EXPLANATION

This activity involves three states of water and two states of carbon dioxide. The carbon dioxide changes from a solid to a gas and bubbles out of the water. As it is denser than air it accumulates near the floor of the room. The carbon dioxide gas is very cold, so it cools the air around it, causing water vapour to condense into droplets. Meanwhile, some of the liquid water in the beaker will have turned into ice as it was frozen by the dry ice.

**A whale's spout is not a stream of water — it is just air that has been blown out of its lungs. The air is warm and moist from the lungs and when it hits the cooler, outer air the water vapour in it condenses into fog.**

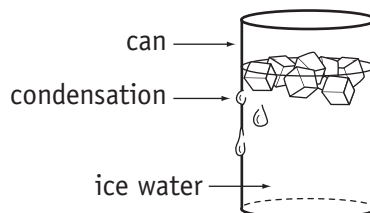
**Any known substance that can be heated to its boiling point will evaporate. Iron evaporates at about 2900°C.**

### HISTORICAL NOTE

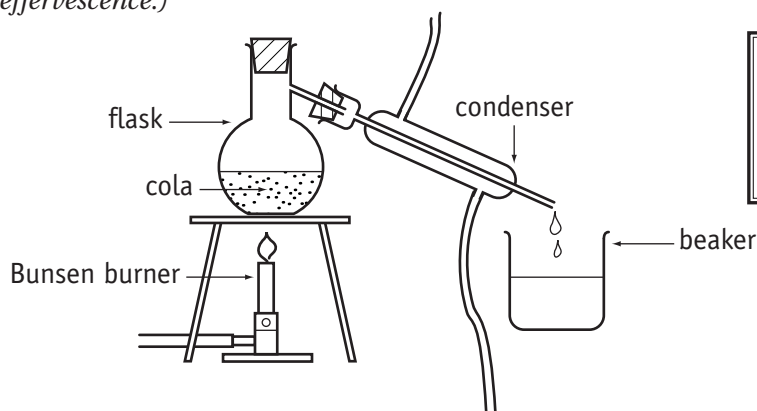
In 1970 one storm in Kansas, USA, reportedly produced hailstones the size of bowling balls. Miraculously no-one was killed.

## LINKED ACTIVITIES

- 1 **Does this can leak?** Fill a tin can with icewater. Observe the outside of the can. *(The cold metal cools the air in contact with it. As the air cools, the water vapour in it condenses and will appear as drops on the outside of the can.)*



- 2 **Change the state of cola.** *TEACHER DEMONSTRATION ONLY — Safety glasses and laboratory coats should be worn.* Pour half a can of cola into a flask and attach it to a condenser. Predict what sort of liquid will be collected from the condenser when the cola is boiled. *(Cola is about 90% water by volume, which will be collected by evaporation and condensation. A can of cola also contains, among other things, about 6 teaspoons of sugar, kola nut extracts for flavour and carbon dioxide for effervescence.)*



### HISTORICAL NOTE

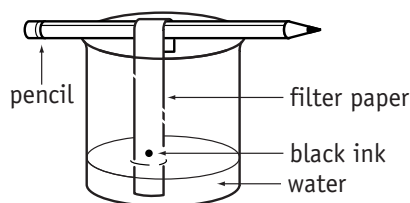
Up until 1905, some types of cola contained cocaine.

# INK SPOTS

You can make other colours come out of a spot of black ink.

## HOW TO DO IT

- 1 Cut a thin strip of filter paper about 15 cm long.
- 2 Use a water soluble felt pen to put a spot of black ink about 2 cm from one end. Dab the pen several times on the spot so there is plenty of ink on it.
- 3 Position the end of the filter paper in some water in a beaker but make sure the ink spot is not in the water. *(The ink will move upwards and various colours will appear.)*



## EXPLANATION

Black ink consists of a mixture of several different-coloured pigments. These pigments have differing solubilities, and the more soluble ones are carried further up the filter paper by the water.

## TO INVESTIGATE

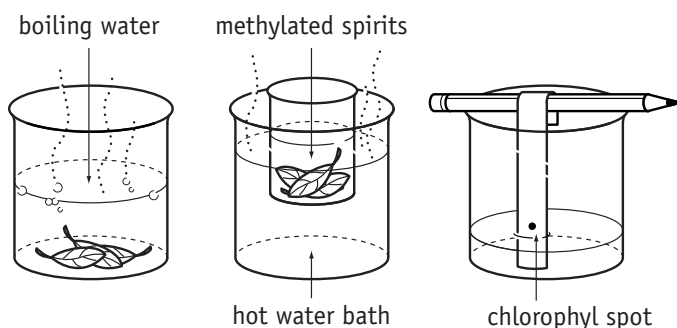
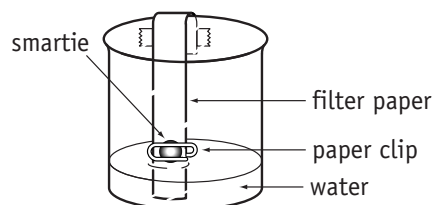
What colours would you find in green, pink, orange and brown water-soluble felt pens? Do all black inks contain the same pigments?

**Some pigments can take on more than one colour. Hydrangeas can be pink or blue, depending on the acidity of the soil. It is the pigment that changes colour in response to pH.**

**The primary colours of inks and paints are red, blue and yellow, but the primary colours of light are red, blue and green.**

## LINKED ACTIVITIES

- 1 **Separate the pigments in Smarties.** Use a paperclip to attach a brown Smartie to a strip of filter paper, close to one end. Dangle the end of the filter paper in water and tape the other end to the outside of the beaker. *(The food colouring will dissolve in the water and the pigments will separate.)*
- 2 **Separate the pigments in chlorophyll.** **TEACHER DEMONSTRATION ONLY** — *Safety glasses and laboratory coat should be worn.* Place a few soft green leaves in boiling water for 2 minutes, then transfer them to a small beaker of methylated spirits in a hot water bath (in a fume cupboard) for about 20 minutes. Put 1 drop of the resulting green solution near the end of a strip of filter paper and let it dry. Repeat this several times until there is a strong green spot. Place the end of the strip in methylated spirits in a fume cupboard. *(Chlorophyll is a mixture of at least three pigments. Green and yellow pigments will be observed on the filter paper.)*



**CREATIVE LITERACY** The Russian botanist, Mikhail Semyonovich Tsvet, developed the process of chromatography and discovered the pigments in chlorophyll. Write a speech that he could have used to announce his discovery to the world press.

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## INVISIBLE INK

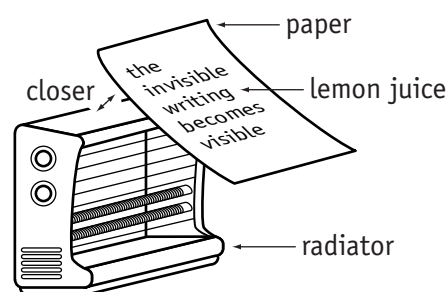
This ink will only appear when it is heated.

## HOW TO DO IT

## TEACHER DEMONSTRATION ONLY

*Safety warning: this activity requires a safe method of disposing of sheets of paper that have accidentally ignited.*

- 1 Dip the wooden end of a match stick into some lemon juice then write with it onto a sheet of paper.
- 2 Let the lemon juice dry on the paper. (*The writing will be almost invisible.*)
- 3 Hold the sheet of paper very close to an electric radiator, but not so close that the paper burns. (*The lemon juice will darken and become visible.*)



## EXPLANATION

The lemon juice contains sugar and other carbon compounds. Upon heating, these begin to break down and some of the carbon appears by itself. Pure carbon is black, so the colour of the lemon juice slowly gets darker as more and more carbon appears.

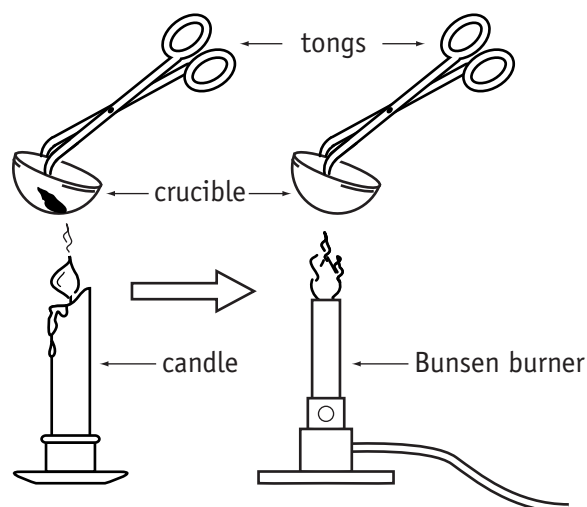
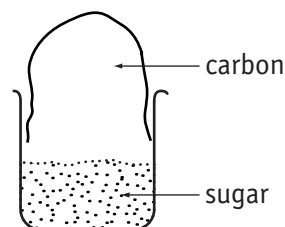
**The charcoal on barbecued meat contains a substance that causes cancer.**

## HISTORICAL NOTE

Some early scientists believed that the sun was a large mass of burning coal.

## LINKED ACTIVITIES

- 1 **Make carbon from sugar.** *TEACHER DEMONSTRATION ONLY* This activity must be carried out in a fume cupboard. Safety glasses, laboratory coat and acid-proof gloves should be worn. Add powdered sugar to a 200 mL beaker until it is about half full. Pour in about 10 mL of concentrated sulfuric acid. (*The carbon will become separated from the hydrogen and oxygen in the sugar and, within a few minutes a block of carbon will rise up from the beaker. Sulfur dioxide fumes will also be produced.*)
- 2 **Make carbon disappear.** Safety glasses and laboratory coats should be worn. Use tongs to hold a white crucible in a candle flame for a moment, so that a light streak of soot is deposited on its outer surface. Then position it above a blue Bunsen burner flame. (*The soot will slowly disappear as the hotter flame causes the complete combustion of the carbon.*)



**CREATIVE LITERACY** Write a series of steps for making caramel. Add notes to explain what is happening to the sugar.

## FOOL'S GOLD

It's easy to tell the difference between gold and fool's gold.

## HOW TO DO IT

- 1 Rub a piece of white paper on a gold ring. *(The gold will not leave a mark on the paper.)*
- 2 Rub a piece of white paper on some fool's gold (iron pyrites). *(A black mark will appear on the paper.)*

## EXPLANATION

Iron pyrites ( $\text{FeS}_2$ ) is a compound containing iron and sulfur, so its properties are quite different to those of gold. This activity is an example of a streak test, in which the colour of a streak of the mineral can be used to help identify it.

**CREATIVE LITERACY** You are an atom of gold. Explain the properties that make you so much better than any of the other elements.



## HISTORICAL NOTE

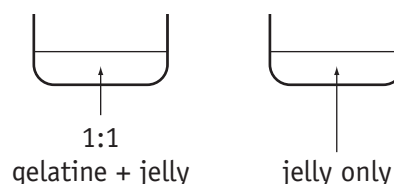
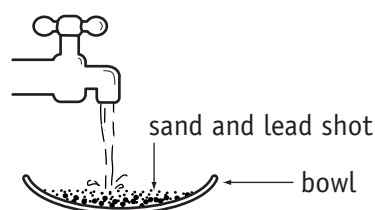
Before the 17th century, most chemists devoted their lives to trying to make gold by mixing different substances together. Some of their mixtures contained mercury, iron ore, coal, egg yolk and even horse manure.

There is no gold in our Australian 'gold' coins. One dollar and two dollar coins are 92% copper, 6% aluminium and 2% nickel.

There are hundreds of kilograms of gold in each cubic kilometre of seawater, but we have not yet found a profitable way to extract it.

## LINKED ACTIVITIES

- 1 **Model gold panning.** Place a mixture of sand and lead shot in a bowl. Run some tapwater into the bowl. Tip the bowl slightly to one side and swirl it so that some of the water and sand leaves the bowl. Repeat this a few times until there is only lead shot in the bowl. *(Lead shot, like gold, is more dense than sand, so it sinks more rapidly in the moving water and will thus remain in the bowl.)*
- 2 **Model the carat in gold rings.** *(Twenty-four carat gold is pure. Other carat numbers indicate the proportion of gold out of a total of 24. Eighteen carat gold is 18 parts gold and 6 parts copper; 12 carat gold is half gold and half copper. The copper is added to make the gold harder.)* Use lemon jelly mixture to represent gold and gelatine to represent copper. For example, a 12 carat mixture would be half jelly mixture and half gelatine; pure gold would be jelly mixture only. *(The more gelatine in the mixture the harder the jelly will become.)*

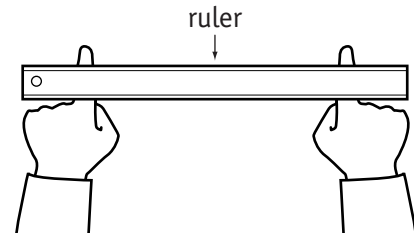


## BALANCE A RULER

No matter how hard you try, you can only do this one hand at a time.

### HOW TO DO IT

- 1 Place a wooden ruler on top of your two hands so the ends of the ruler are supported by your two index fingers.
- 2 Slowly slide your two index fingers towards the centre of the ruler. (*Only one finger at a time will move along the ruler. The fingers will alternate until they meet at the balance point of the ruler.*)
- 3 Now try to move both fingers back to their original positions. (*Only one finger will move along the ruler.*)



### EXPLANATION

The moving finger is whichever one has the least weight on it and, consequently, the least friction. At any one time, the amount of weight on each finger will depend on its location relative to the ruler's centre of gravity and the turning moments at each finger.

### TO INVESTIGATE

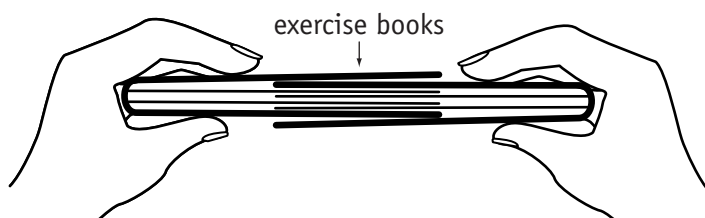
Predict what would happen if you used an irregular object, such as an axe handle or a pool cue, instead of a ruler.

**Railway lines are very smooth, but they still do their work by friction. The train engines are so heavy that there is enough friction between the wheel and the tracks to provide grip.**

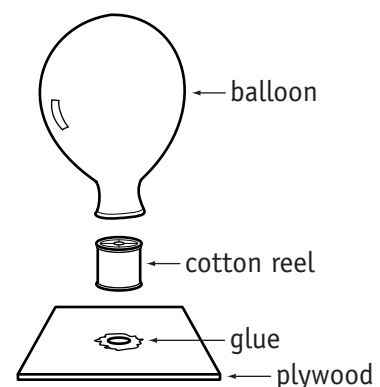
**Friction heat ignites matches.**

### LINKED ACTIVITIES

- 1 **Feel the force of friction.** Take two exercise books and interlock them by alternating at least 20 pages from each book. Make sure they overlap over at least half their width. Hold the books at the bind and try to pull them apart. (*Friction between the pages will make them very hard to pull apart.*)



- 2 **Make a model hovercraft** using a balloon, a cotton reel, some glue and a small sheet of plywood with a hole in the centre. Blow up the balloon then quickly push the hovercraft over a smooth surface. (*It will seem to glide because the air from the balloon decreases the friction between the plywood and the surface below.*)



**CREATIVE LITERACY** Choose a sport and write a short list identifying all the times when friction occurs.

## 37

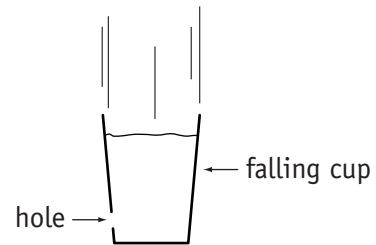
# FALLING CUP

A cup with a hole in it will lose water, but not if it is in free fall.

## HOW TO DO IT

This activity should be done outside.

- 1 Fill a polystyrene cup with water. Use a pen to put a hole near the bottom of the cup. (The water will drain out.)
- 2 Place your finger over the hole and fill the cup again.
- 3 Hold the cup up high and let it drop to the ground. (*The water will not come out of the hole while the cup is falling.*)



## EXPLANATION

The force of gravity makes the water drain from the cup when it is held in the air. When the cup is falling, both the water and the cup are accelerating at the same rate so the water is apparently weightless in comparison to the cup.

**Astronauts train for weightlessness in space by moving around and 'floating' inside a diving aeroplane (affectionately referred to as the 'vomit comet'). The apparent weightlessness that space shuttle astronauts experience in space is actually caused by the same phenomenon — the craft is continuously falling in a curved path around the earth.**

**In 1944, an airman fell out of a Lancaster bomber at an altitude of 5 km and lived.**

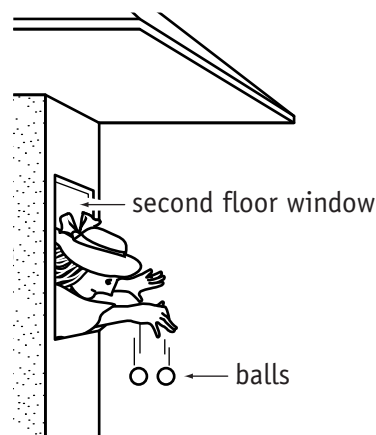
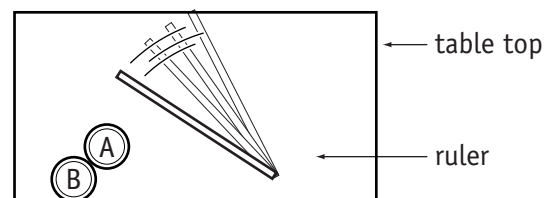
**If a bullet is fired from a gun that is horizontal, it will hit the ground at the same time as another bullet which is simply dropped from that height.**

### HISTORICAL NOTE

When Galileo proved that the earth is not the centre of the universe, he was brought before the Inquisition, convicted as a heretic and forced to spend his remaining years (8 years, in fact) under house arrest.

## LINKED ACTIVITIES

- 1 **Does the path of an object influence the time it takes to fall to the ground?** Place two 5c coins, touching each other, near the edge of a table in the positions shown in the diagram. Flick coin 'A' with a ruler, so it is propelled across the room, and coin 'B' will fall more directly to the floor (you will hear the sound better on a wooden floor). (*They will both hit the ground at the same time because gravity acts equally on both coins.*)
- 2 **Did Galileo drop balls from the Tower of Pisa?** Drop a ping-pong ball and a golf ball at the same time from a second storey window. (*This is a rough approximation because the balls are different sizes, but the heavy ball will hit the ground first. There is no factual basis for the story of Galileo dropping balls from the Tower of Pisa.*)

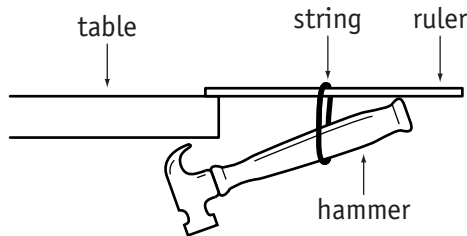


## SUSPENDED RULER

Only a small part of this ruler is on the table, but it still doesn't fall to the ground.

### HOW TO DO IT

- 1 Use a length of string to loosely tie a ruler to a hammer.
- 2 Position the end of the ruler on a table top, so the head of the hammer is suspended under the table.  
(The ruler will not fall.)



### EXPLANATION

An object is balanced if its centre of gravity is on, or just below, the point where it is supported. In this case, the head of the hammer is very heavy, so the centre of gravity of the system is somewhere below the edge of the table.

**Scientists still don't know what causes gravity.**

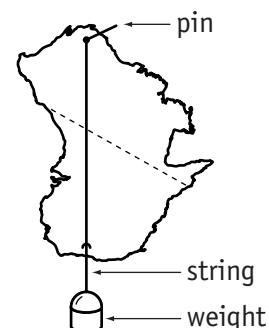
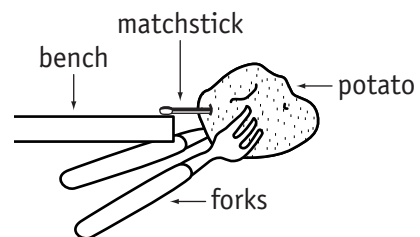
**The Leaning Tower of Pisa is in genuine danger of collapsing, but it has not yet been decided how to save it. It leans over 5 m off centre.**

### HISTORICAL NOTE

Ships occasionally tip over if their centre of gravity is too high. One of the most famous examples of this was the *Normandie*, a large ocean liner, which caught fire while docked in New York Harbour in 1942. The firemen standing on the dock hosed so much water into the upper layers of the ship that it capsized and had to be scrapped.

### LINKED ACTIVITIES

- 1 **Suspend a potato.** Insert a matchstick and two forks (with heavy handles) into a potato so that it is suspended from the corner of a bench, as shown in the diagram. (The heavy handles of the forks cause the centre of gravity to be located near or under the bench top.)
- 2 **Find the centre of gravity of Australia.** Cut out a cardboard shape of Australia, hang it from a pin, and mark the direction of a plumb bob hanging from the same pin. Move the pin to another part of the shape and mark the direction again. The centre of gravity is where the two lines meet. Which town is closest to it? (Australia doesn't really have a centre of gravity, but this technique can be used to find the centre of gravity of an irregularly shaped object.)

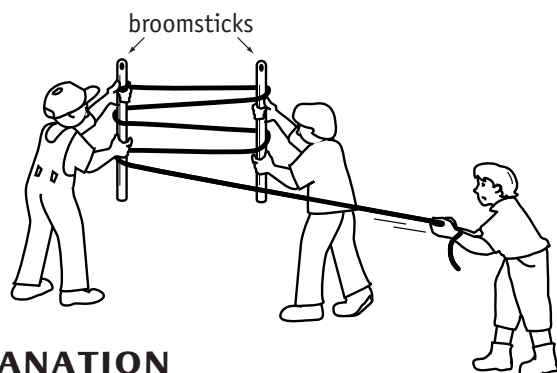


## BROOMSTICKS

One student can exert more force than two students.

### HOW TO DO IT

- 1 Get two people to each hold a broomstick, about 1 m apart.
- 2 Tie the end of a rope to one of the sticks, then wind the rope three times around both sticks, as shown in the diagram.
- 3 Ask a third person to pull the rope while the other two try to stay where they are. *Safety warning: fingers should be kept clear of the rope. (The two broomsticks will be pulled together.)*



### EXPLANATION

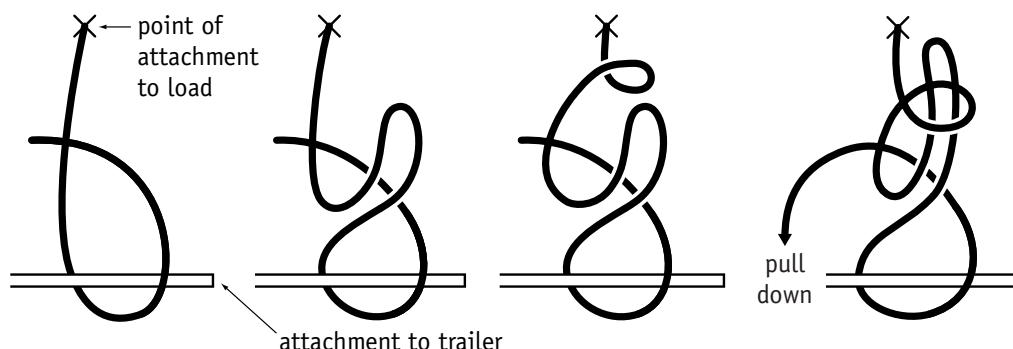
The sticks and rope are like a simple block and tackle. A block and tackle with four pulleys uses one quarter of the force normally needed to lift the object. However, in the activity above, friction will obviously reduce the efficiency of the system.

### TO INVESTIGATE

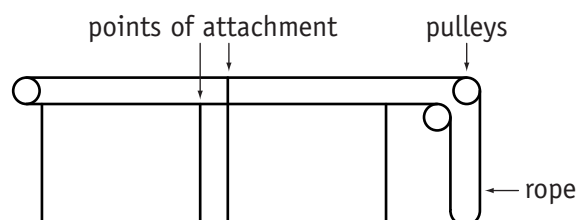
Use laboratory pulleys to make a block and tackle that requires one quarter of the force normally needed to lift a small bucket of sand.

### LINKED ACTIVITIES

- 1 Tie a truckie's knot, as shown in the diagram. (This is a block and tackle completely made of rope. It is useful for tying down loads on trucks and trailers.)



- 2 Solve the following problem. Design a pulley system that could operate theatre curtains. (Pulleys can also be used to change the direction of a force.)



### HISTORICAL NOTE

Archimedes invented the block and tackle over 200 years before the birth of Christ. He demonstrated its effectiveness when, with no other assistance, he pulled a ship up onto dry land.

### HISTORICAL NOTE

In Stonehenge (England), there are huge blocks of stone which have been placed high on top of other huge blocks of stone. Engineers have no idea how they could have been lifted up

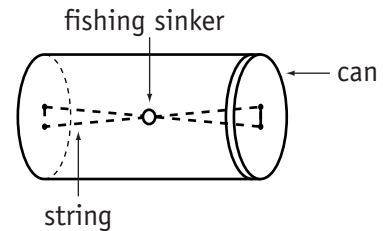
## 40

# RETURNING CAN

If you roll this can away, it will come back to you.

## HOW TO DO IT

- 1 Obtain a can with a lid. Put two nail holes about 1 cm apart on the base of the can, close to the centre. Also do this with the lid.
- 2 Connect the holes with string as shown in the diagram.
- 3 Attach a heavy fishing sinker to the centre of the string.
- 4 Gently roll the can on a hard, level surface. *(The can will roll to a stop about 1 m away and then roll back to you.)*



## EXPLANATION

As the can rolls away from you, the lead weight does not roll, so the string winds up. In this way, kinetic energy is changed to potential energy, which is stored as tension in the string and a slightly higher position of the weight. When the can rolls back, the potential energy is converted to kinetic energy.

**Window glass is highly elastic over short distances because it always springs back to exactly where it was.**

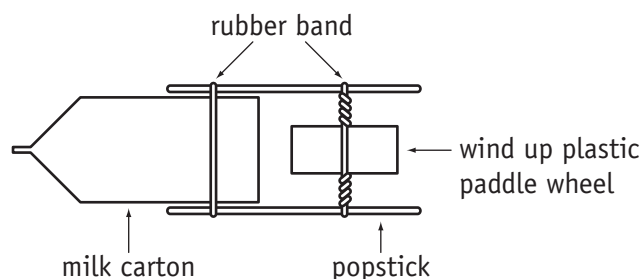
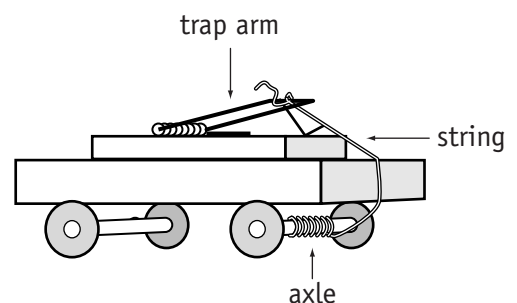
**CREATIVE LITERACY** Carefully pull apart an old alarm clock or clockwork toy. Prepare an oral presentation describing where the potential energy is stored and how it is connected to moving parts.

## HISTORICAL NOTE

When you give someone the 'two fingers' you are using a signal that has its origins in elastic potential energy. The first two fingers of your right hand are the two fingers used to pull back the drawstring of a bow (a device that uses elastic potential energy). The two-fingered salute originated in the Middle Ages, when English bowmen would hold up their two fingers to taunt their enemy before the battle began. However, in reprisal, the French would summarily remove these two fingers from any bowmen who were unlucky enough to be captured.

## LINKED ACTIVITIES

- 1 **Make a mouse trap car.** Attach one end of a length of string to the spring lever of the mouse trap, and wind the other end around the front axle of the car. *(As the trap closes, the potential energy in the spring is converted to kinetic energy. This movement pulls the string, which is wound around the axle, so the car will move forward.)*
- 2 **Solve the following problem.** Use elastic potential energy to make a milk carton boat move across a tub of water. *(One possible solution is shown.)*



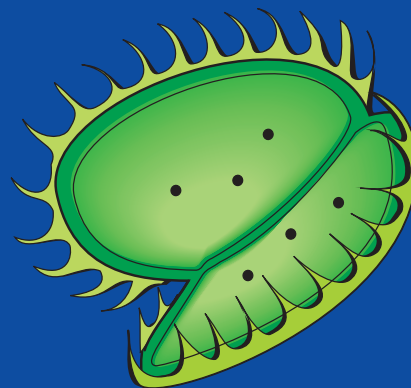
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